

Lenovo ThinkSystem SC777 V4 Neptune Server

Product Guide

The ThinkSystem SC777 V4 Neptune node is the next-generation high-performance accelerated server based on the sixth generation Lenovo Neptune® direct water cooling platform.

Based on the NVIDIA GB200 NVL4 platform, the ThinkSystem SC777 V4 server combines the latest high-performance NVIDIA GPU and Lenovo's market-leading water-cooling solution, which results in extreme HPC/AI performance in dense packaging.

The direct water cooled solution is designed to operate by using warm water, up to 45°C (113°F). Chillers are not needed for most customers, meaning even greater savings and a lower total cost of ownership. Eight servers are housed in the ThinkSystem N1380 enclosure, a 13U rack mount unit that fits in a standard 19-inch rack.

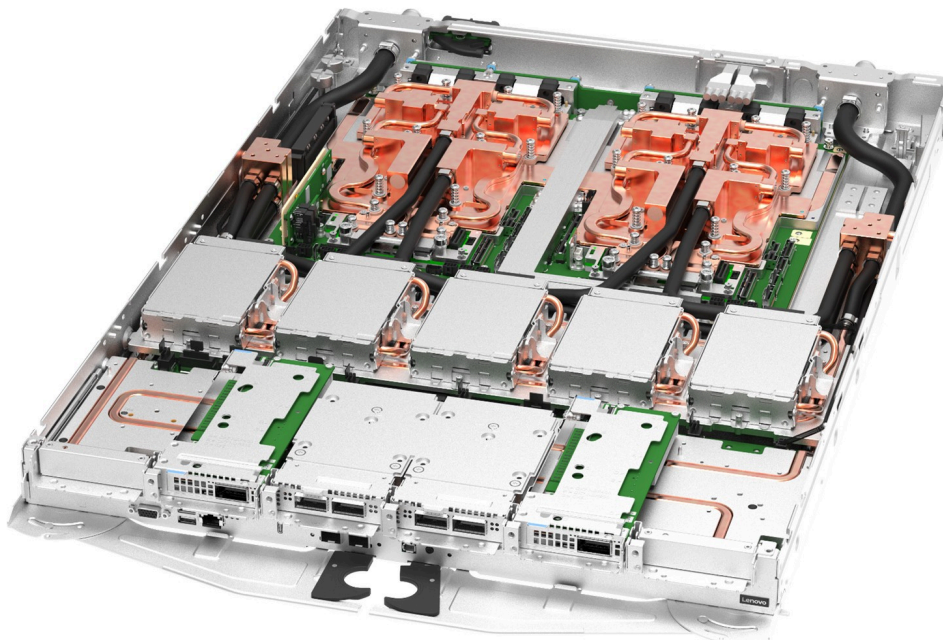


Figure 1. The Lenovo ThinkSystem SC777 V4 Neptune Server (AI Training configuration)

[360° View](#)
[Full 3D Tour](#)

Did you know?

Lenovo Neptune and the ThinkSystem SC777 V4 are designed to operate at water inlet temperatures as low as the dew point allows, up to 45°C. This design eliminates the need for additional chilling and allows for efficient reuse of the generated heat energy in building heating or adsorption cold water generation, for example. It uses treated de-ionized water as a coolant, a much safer and more efficient alternative to the commonly used PG25 Glycol liquid, reflecting our commitment to environmental responsibility.

Key features

The ThinkSystem SC777 V4 server uses a new design, where the servers are mounted vertically in the N1380 enclosure. Eight servers are installed in one 13U N1380 enclosure.

Revolutionary Lenovo Neptune Design

Engineered for large-scale Artificial Intelligence (AI) and High Performance Computing (HPC), Lenovo ThinkSystem SC777 V4 Neptune excels in intensive simulations and complex modeling. It is designed to handle modeling, training, simulation, rendering, financial tech, scientific research, technical computing, grid deployments, and analytics workloads in various fields such as research, life sciences, energy, engineering, and financial simulation.

At its core, Lenovo Neptune applies 100% direct warm-water cooling, maximizing performance and energy efficiency without sacrificing accessibility or serviceability. The SC777 V4 integrates seamlessly into a standard 19" rack cabinet with the ThinkSystem N1380 Neptune enclosure, featuring a patented blind-mate stainless steel dripless quick connection.

This design ensures easy serviceability and extreme performance density, making the SC777 V4 the go-to choice for compute clusters of all sizes - from departmental/workgroup levels to the world's most powerful supercomputers – from Exascale to Everscale®.

Lenovo Neptune utilizes superior materials, including custom copper water loops and patented CPU cold plates, for full system water-cooling. Unlike systems that use low-quality FEP plastic, Neptune features durable stainless steel and reliable EPDM hoses. The N1380 enclosure features an integrated manifold that offers a patented blind-mate mechanism with aerospace-grade drip-less connectors to the compute trays, ensuring safe and seamless operation.

Lenovo's direct water-cooled solutions are factory-integrated and are re-tested at the rack-level to ensure that a rack can be directly deployed at the customer site. This careful and consistent quality testing has been developed as a result of over a decade of experience designing and deploying DWC solutions to the very highest standards.

Scalability and performance

The ThinkSystem SC777 V4 server and ThinkSystem N1380 enclosure offer the following features to boost performance, improve scalability, and reduce costs:

- The SC777 V4 node leverages the NVIDIA GB200 NVL4 architecture:
 - Two NVIDIA Grace processors
 - Four NVIDIA Blackwell B200 GPU modules
 - Highspeed NVLink interconnects between GPUs
- The server has up to 6x PCIe Gen 5 x16 slots for high-speed networking, depending on the configuration.
- The server also supports up to 10x E3.S 1T drive bays, depending on the configuration, for internal storage and OS boot
- All supported drives are E3.S NVMe drives with PCIe Gen 5 x4 host interface, to maximize I/O performance in terms of throughput, bandwidth, and latency.
- The node includes two 25 Gb Ethernet onboard ports for cost effective networking. High speed host networking can be added through the included PCIe slots. One of the 25Gb ports support NC-SI for BMC management communication to the BMC in addition to the host communication.
- The node offers PCI Express 5.0 (PCIe Gen 5) I/O expansion capabilities that doubles the theoretical maximum bandwidth of PCIe Gen 4 (32GT/s in each direction for Gen 5, compared to 16 GT/s with Gen 4). A PCIe Gen 5 x16 slot provides 128 GB/s bandwidth, enough to support a 400GbE network connection. Two Gen 5 x16 connections (32 lanes total) provides the bandwidth needed for an 800 GbE connection using the ConnectX-8 8180 network adapter.

Energy efficiency

The direct water cooled solution offers the following energy efficiency features to save energy, reduce operational costs, increase energy availability, and contribute to a green environment:

- Water cooling eliminates power that is drawn by cooling fans in the enclosure and dramatically reduces the required air movement in the server room, which also saves power. In combination with an Energy Aware Runtime environment, savings as much as 40% are possible in the data center due to the reduced need for air conditioning.
- Water chillers may not be required with a direct water cooled solution. Chillers are a major expense for most geographies and can be reduced or even eliminated because the water temperature can now be 45°C instead of 18°C in an air-cooled environment.
- With the new water-cooled power conversion stations and SMM3, essentially 100% system heat recovery is possible, depending on water and ambient temperature chosen. At 45°C water temperature and 30°C room temperature it will be typically around 95% through surface radiated heat. Heat energy absorbed may be reused for heating buildings in the winter, or generating cold through Adsorption Chillers, for further operating expense savings.
- The processors and other microelectronics are run at lower temperatures because they are water cooled, which uses less power, and allows for higher performance.
- The processors are run at uniform temperatures because they are cooled in parallel loops, which avoid thermal jitter and provides higher and more reliable performance at same power.
- 80 PLUS Titanium-compliant power conversion stations ensure energy efficiency.
- Power monitoring and management capabilities through the System Management Module in the N1380 enclosure.
- The System Management Module (SMM3) automatically calculates the power boundaries at the enclosure level; the protective capping feature ensures overall power consumption stay within limits, adjusting dynamically based on the server and power conversion station status change.
- Lenovo power/energy meter based on TI INA228 measures DC power for the SC777 V4 at higher than 97% accuracy and up to 100 Hz sampling frequency to the XCC and can be leveraged both in-band and out-of-band using IPMI raw commands.
- Optional Energy Aware Runtime provides sophisticated power monitoring and energy optimization on a job-level during the application runtime without impacting performance negatively.

Manageability and security

The following powerful systems management features simplify local and remote management of the SC777 V4 server:

- The server includes an XClarity Controller 3 (XCC3) to monitor server availability. XCC3 Premier is an included offering to provide remote control (keyboard video mouse) functions, support for the mounting of remote media files, FIPS 140-3 security, enhanced NIST 800-193 support, boot capture, power capping, and other management and security features.
- Lenovo XClarity One offers comprehensive hardware management tools that help to increase uptime, reduce costs, and improve productivity through advanced server management capabilities.
- Support for industry standard management protocols, IPMI 2.0, SNMP 3.0, Redfish REST API, serial console via IPMI
- The SC777 V4 is enabled with Lenovo HPC & AI Software Stack, so, you can support multiple users and scale within a single cluster environment.
- Lenovo HPC & AI Software Stack provides our HPC customers you with a fully tested and supported open-source software stack to enable your administrators and users with for the most effective and environmentally sustainable consumption of Lenovo supercomputing capabilities.

- Our Confluent management system provides an interface designed to abstract the users from the complexity of HPC cluster orchestration and AI workloads management, making open-source HPC software consumable for every customer.
- Integrated Trusted Platform Module (TPM) 2.0 support enables advanced cryptographic functionality, such as digital signatures and remote attestation.
- Supports Secure Boot to ensure only a digitally signed operating system can be used.
- Industry-standard Advanced Encryption Standard (AES) NI support for faster, stronger encryption.
- With the System Management Module (SMM) installed in the enclosure, only one Ethernet connection is needed to provide remote systems management functions for all SC777 V4 servers and the enclosure.
- The SMM3 management module has two Ethernet ports which allows a single Ethernet connection to be daisy chained, up to 99 connections (eg 3 enclosures and 96 servers), thereby significantly reducing the number of Ethernet switch ports needed to manage an entire rack of SC777 V4 servers and N1380 enclosures.
- The N1380 enclosure and its power conversion stations include drip sensors that monitor the inlet and outlet manifold quick connect couplers; leaks are reported via the SMM.

Availability and serviceability

The SC777 V4 node and N1380 enclosure provide the following features to simplify serviceability and increase system uptime:

- Designed to run 24 hours a day, 7 days a week
- With full water cooling, system fans are not required. This results in significantly reduced noise levels on the data center floor, a significant benefit to personnel having to work on site.
- N+1 redundancy of power conversion stations, with oversubscription, which means greater system uptime.
- Power conversion stations are hot-swappable to minimize downtime.
- Toolless cover removal on the nodes provides easy access to upgrades and serviceable parts such as adapters and drives.
- The XCC offers optional remote management capability and can enable remote keyboard, video, and mouse (KVM) control and remote media for the node.
- Built-in diagnostics in UEFI, using Lenovo XClarity Provisioning Manager, speed up troubleshooting tasks to reduce service time.
- Lenovo XClarity Provisioning Manager supports diagnostics and can save service data to a USB key drive or remote CIFS share folder for troubleshooting and reduce service time.
- Auto restart in the event of a momentary loss of AC power (based on power policy setting in the XClarity Controller service processor)
- Virtual reseal is a supported feature of the System Management Module which, in essence, removes the node from A/C power and reconnecting the node to AC power from a remote location.
- There is a three-year customer replaceable unit and onsite limited warranty, with next business day 9x5 coverage. Optional warranty upgrades and extensions are available.

Performance benchmarks

At Lenovo, we know customers care most about how their workloads perform in real-world environments, which is why we carefully tune our server hardware, firmware, and drivers to help deliver strong benchmark results that reflect practical business needs. The goal is simple: to help customers run their critical applications more efficiently and achieve better outcomes.

Lenovo performance engineers have achieved world record performance benchmarks on many ThinkSystem servers. See the complete list of current world records in the Lenovo Press article [Performance Benchmark World Record Summary for Lenovo ThinkSystem Servers](#).

The Lenovo Server Performance Team also publishes reports on key performance world records at the time the world record is announced. See all published [Performance Benchmark Reports](#).

View in Augmented Reality

View the SC777 V4 in augmented reality (AR) using your smartphone or tablet.

Simply follow these steps:

1. Scan the QR code* with the camera app on your phone
2. Point your phone at a flat surface
3. Wait a few seconds for the model to appear

Once the server appears, you can move your phone around it. You can also drag or rotate the server to reposition it.



For more information about the AR viewer, see the article "Introducing the Augmented Reality Viewer for Lenovo Servers", available from <https://lenovopress.lenovo.com/lp1952>

* If you're viewing this document on your phone or tablet, simply tap the QR code

Components and connectors

The front of the server is shown in the following figure.

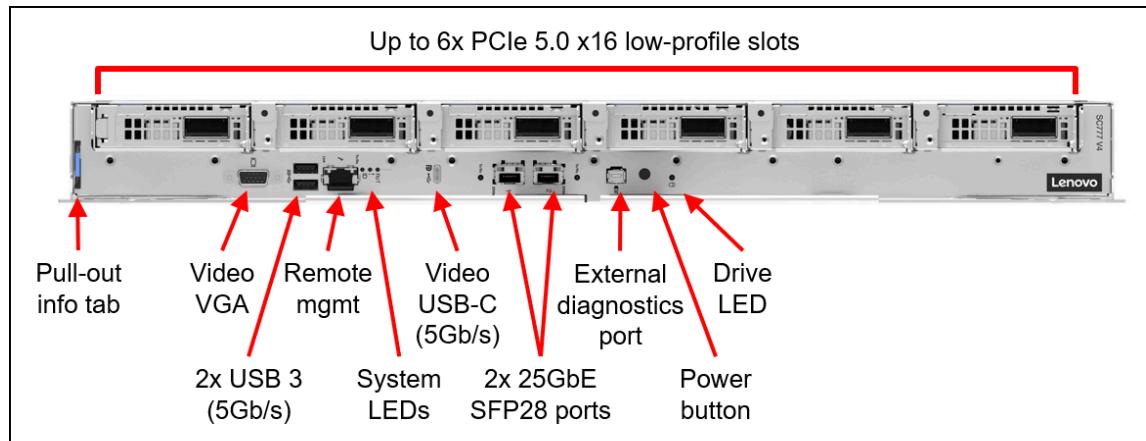


Figure 2. Front view of the ThinkSystem SC777 V4

The following figure shows key components internal to the server.

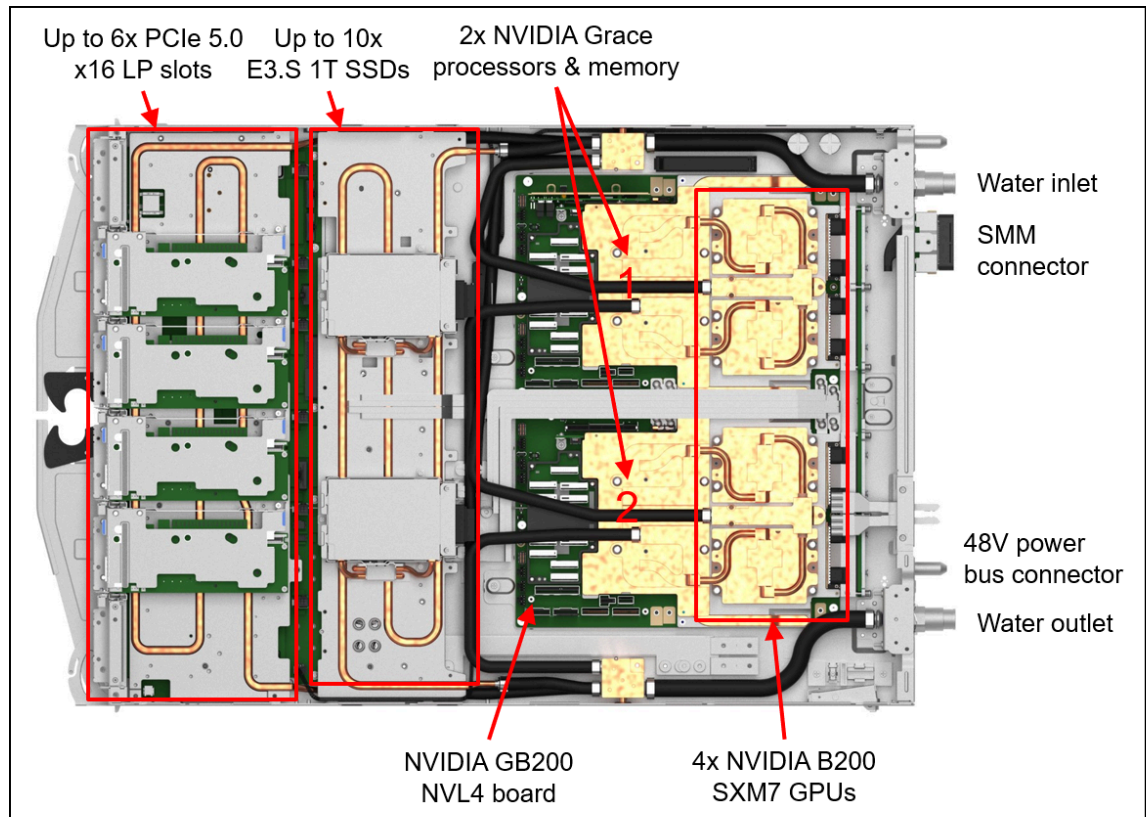


Figure 3. Inside view of the SC777 V4 server in the water-cooled tray

The compute nodes are installed vertically in the ThinkSystem N1380 enclosure, as shown in the following figure.

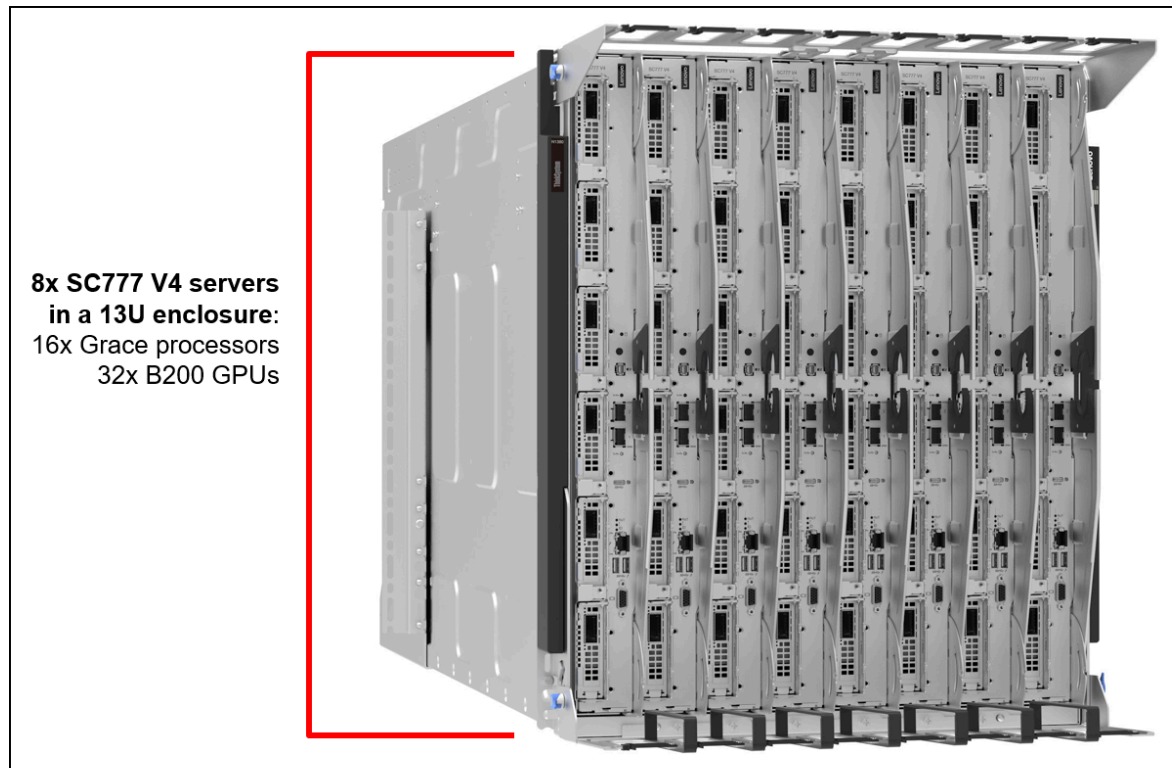


Figure 4. Front view of the N1380 enclosure

The rear of the N1380 enclosure contains the 4x water-cooled power conversion stations (PCS), water connections, and the System Management Module.

Note: The short hoses that are attached to the water inlet and water outlet have been removed from the figure for clarity.

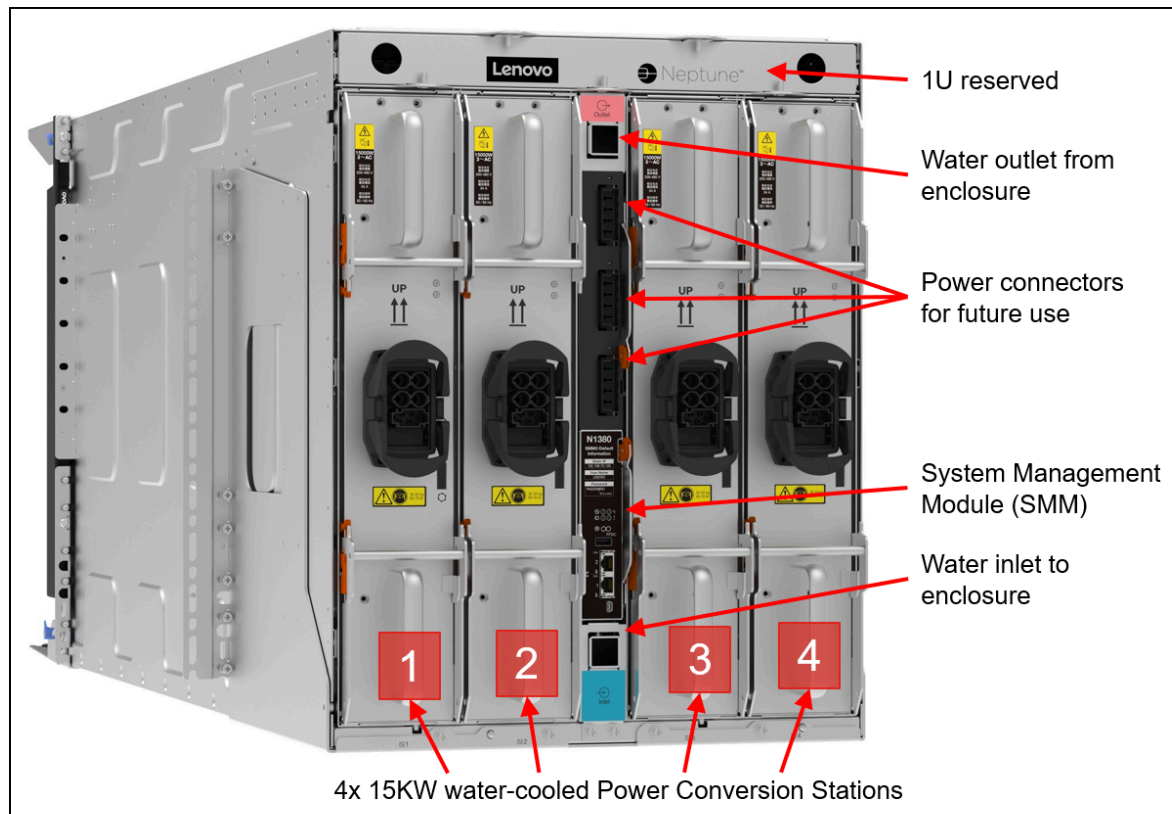


Figure 5. Rear view of the N1380 enclosure (water hoses removed for clarity)

System architecture

The SC777 V4 is offered in two primary configurations:

- HPC and AI for Science workloads
- AI Training workloads

The following figure shows the architectural block diagram of the SC777 V4 best suited to HPC and AI for Science workloads. This configuration offers 4x GPUs and 6x PCIe Gen5 x16 slots all directly connected to the CPUs. This configuration optimizes network connectivity while maximizing computational efficiency, ensuring cost-effectiveness in performance.

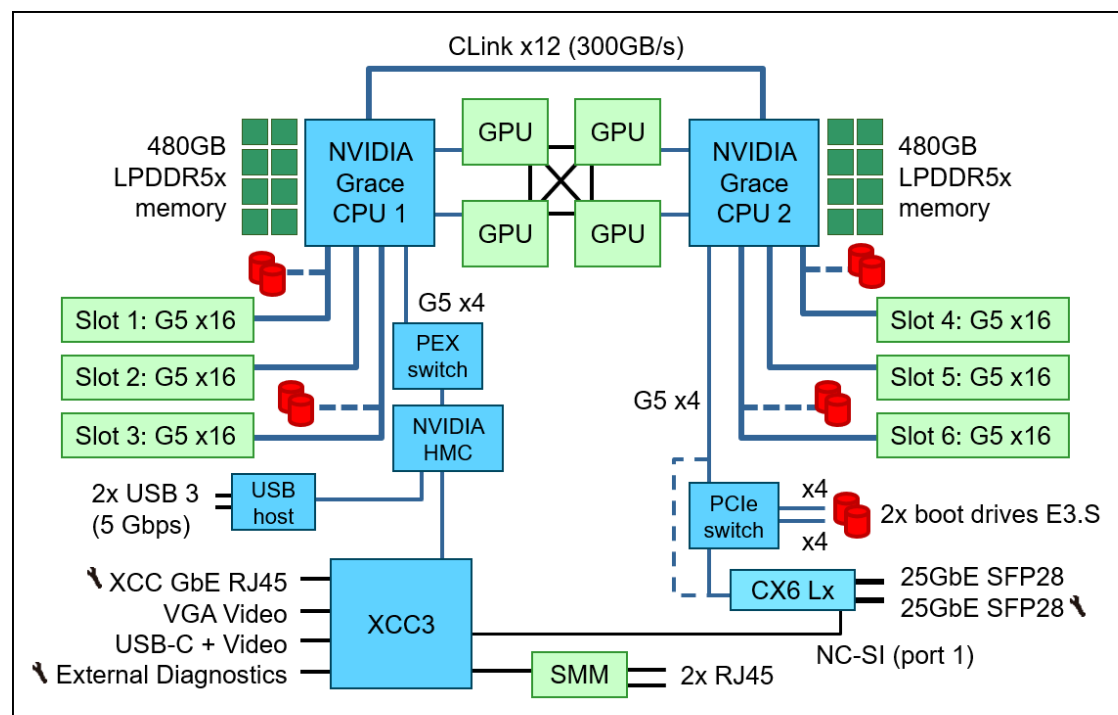


Figure 6. SC777 V4 system architectural block diagram - HPC and AI for Science workloads

The following figure shows the architectural block diagram of the SC777 V4 best suited to AI Training workloads. This configuration implements GPUDirect by connecting the GPUs, 800Gb/s OSFP network interfaces and drives all together using ConnectX-8 switches. The AI Training configuration is optimized for scale-out GPU performance which provides advantages in large AI training workloads.

Standard specifications - SC777 V4 tray

The following table lists the standard specifications of the SC777 V4 server tray.

Table 1. Standard specifications - SC777 V4 tray

Components	Specification
Machine type	7DMP - 3-year Base Warranty 7DKA - 5-year Warranty-Ready
Form factor	NVIDIA GB200-NVL4 architecture system in a water-cooled server tray, installed vertically in an enclosure
Enclosure support	ThinkSystem N1380 enclosure
Processor	Two NVIDIA Grace processors, with 72 Arm Neoverse V2 cores, core speeds of up to 3.1 GHz base frequency, and TDP rating of 300W. Supports PCIe 5.0 for high performance connectivity to network adapters and NVMe drives.
Chipset	None. Integrated into the processor.
GPUs	Four NVIDIA Blackwell B200 GPUs per tray, each with 186 GB HBM3e memory
CPU memory	Up to 480GB of LPDDR5 ECC memory per processor, memory operates at 4266 MHz
Memory maximum	960GB per node
Memory protection	ECC with Single Error Correction (SEC), Double Error Detection (DED, parity, scrubbing, poisoning, and page off-lining)
Disk drive bays	Each node supports up to 10x EDSFF E3.S 1T NVMe SSDs, installed in 5 internal drive bays (non-hot-swap). Each drive has a PCIe x4 host interface.
Maximum internal storage	153.6TB using 10x 15.36TB E3.S NVMe SSDs
Storage controllers	Onboard NVMe ports (RAID provided by operating system, if desired)
Optical drive bays	None
Network interfaces	2x 25 Gb Ethernet SFP28 onboard connectors based on NVIDIA ConnectX-6 Lx controller (support 10/25Gb) 25Gb port 1 can optionally be shared with the XClarity Controller 3 (XCC3) management processor for Wake-on-LAN and NC-SI support.
PCIe slots	Up to 6x PCIe 5.0 x16 slots supporting low-profile PCIe adapters
Ports	External diagnostics port, VGA video port, USB-C port for both video (DisplayPort) and data (5 Gb/s), 2x USB 3 (5 Gb/s) ports, RJ-45 1GbE systems management port for XCC remote management. Additional ports provided by the enclosure as described in the Enclosure specifications section.
Video	Embedded graphics with 16 MB memory with 2D hardware accelerator, integrated into the XClarity Controller 3 management controller. Two video ports (VGA port and USB-C DisplayPort video port); both can be used simultaneously if desired. Maximum resolution is 1920x1200 32bpp at 60Hz.
Security features	Power-on password, administrator's password, Trusted Platform Module (TPM), supporting TPM 2.0.

Components	Specification
Systems management	Operator panel with status LEDs. Optional External Diagnostics Handset with LCD display. Clarity Controller 3 (XCC3) embedded management based on the ASPEED AST2600 baseboard management controller (BMC), XClarity One centralized infrastructure delivery. Optional XCC3 Premier to enable remote control functions and other features. Lenovo power/energy meter based on TI INA228 for 100Hz power measurements with >97% accuracy. System Management Module (SMM3) in the N1380 enclosure provides additional systems management functions from power monitoring to liquid leakage detection on chassis, tray and power conversation station level.
Operating systems supported	Ubuntu is Supported & Certified. See the Operating system support section for details and specific versions.
Limited warranty	Three-year or five-year customer-replaceable unit and onsite limited warranty with 9x5 next business day (NBD).
Service and support	Optional service upgrades are available through Lenovo Services: 4-hour or 2-hour response time, 6-hour fix time, 1-year or 2-year warranty extension, software support for Lenovo hardware and some third-party applications.
Dimensions	Width: 546 mm (21.5 inches), height: 53 mm (2.1 inches), depth: 760 mm (29.9 inches)
Weight	36.5 kg (80.5 lbs)

Standard specifications - N1380 enclosure

The ThinkSystem N1380 enclosure provides shared high-power and high-efficiency power conversion stations. The SC777 V4 servers connect to the midplane of the N1380 enclosure. This midplane connection is for power (via a 48V busbar) and control only; the midplane does not provide any I/O connectivity.

The following table lists the standard specifications of the enclosure.

Table 2. Standard specifications: ThinkSystem N1380 enclosure

Components	Specification
Machine type	7DDH - 3-year warranty
Form factor	13U rack-mounted enclosure
Maximum number of SC777 V4 nodes supported	Up to 8x server nodes per enclosure
Node support	ThinkSystem SC750 V4 ThinkSystem SC777 V4
Enclosures per rack	Up to three N1380 enclosures per 42U or 48U rack
System Management Module (SMM)	The hot-swappable System Management Module (SMM3) is the management device for the enclosure. Provides integrated systems management functions and controls the power and cooling features of the enclosure. Provides remote browser and CLI-based user interfaces for remote access via the dedicated Gigabit Ethernet port. Remote access is to both the management functions of the enclosure as well as the XClarity Controller 3 (XCC3) in each node. The SMM has two Ethernet ports which enables a single incoming Ethernet connection to be daisy chained across 3 enclosures and 48 nodes, thereby significantly reducing the number of Ethernet switch ports needed to manage an entire rack of SC777 V4 nodes and enclosures.
Ports	Two RJ45 port on the rear of the enclosure for 10/100/1000 Ethernet connectivity to the SMM3 for power and cooling management.
I/O architecture	None integrated. Use top-of-rack networking and storage switches.
Power supplies	Up to 4x water-cooled hot-swap power conversion stations (PCS), depending on the power policy and the power requirements of the installed server node trays. N+1 redundancy. PCS units are installed at the rear of the enclosure. Single power domain supplies power to all nodes. 80 PLUS Titanium compliant. Built-in overload and surge protection.
Cooling	Direct water cooling supplied by water hoses connected to the rear of the enclosure.
System LEDs	SMM has four LEDs: system error, identification, status, and system power. Each power conversion station (PCS) has AC, DC, and error LEDs. Nodes have more LEDs.
Systems management	Browser-based enclosure management through an Ethernet port on the SMM at the rear of the enclosure. Integrated Ethernet switch provides direct access to the XClarity Controller (XCC) embedded management of the installed nodes. Nodes provide more management features.
Temperature	- Operating water temperature: 2°C to 45°C (35.6°F to 113°F) (ASHRAE S45* compliant) - Operating air temperature: 5°C to 40°C (41°F to 104°F) (ASHRAE A3 compliant) See Operating Environment for more information. * ASHRAE S45, formerly known as ASHRAE W45
Electrical power	380~480Vac, 3-phase WYE
Power cords	Custom power cables for direct data center attachment, 1 dedicated (32A) or 1 shared (63A) per power module

Components	Specification
Limited warranty	Three-year customer-replaceable unit and onsite limited warranty with 9x5/NBD.
Dimensions	Width: 540 mm (21.3 in.), height: 572 mm (22.5 in.), depth: 1302 mm (51.2 in.). See Physical and electrical specifications for details.
Weight	- Empty enclosure (with midplane and cables): 94 kg (208 lbs) - Fully configured enclosure with 4x power conversion stations and 8x SC777 V4 server trays: 470 kg (1036 lbs)

Models

The SC777 V4 node and N1380 enclosure are configured by using the configure-to-order (CTO) process with the Lenovo Cluster Solutions configurator (x-config) or in Lenovo Data Center Solution Configurator (DCSC).

DCSC users: To configure SC777 V4 and N1380 in DCSC, you will need to be in **AI & HPC – ThinkSystem Hardware** mode. General Purpose mode cannot be used to configure these systems.

The following table lists the base CTO models and base feature codes

Table 3. Base CTO models

Machine Type/Model	Feature code	Description
7DMPCTOLWW	C51L	ThinkSystem SC777 V4 Neptune Tray (3-Year Base Warranty)
7DKACTOLWW	C51L	ThinkSystem SC777 V4 Neptune Tray (5-Year Warranty Ready)
7DDHCTOLWW	BYJZ	ThinkSystem N1380 Neptune Enclosure (3-Year Warranty)

The ThinkSystem SC777 V4 node is equipped with high-performance GPUs. It falls under classification 5A992.z per U.S. Government Export regulations, and its market and customer availability are controlled, potentially necessitating an export license.

Configurations

There are two primary configurations offered with the SC777 V4. Block diagrams for these configurations are shown in the [System architecture](#) section.

HPC and AI for Science configuration:

- 4x B200 GPU Modules directly connected to processors with x16 interface (2 to each processor)
- Network connectivity options:
 - Up to 6x ConnectX-7 (single-port 400Gb adapters or dual-port 200Gb adapters), each with a Gen5 x16 connection
 - Up to 3x ConnectX-8 (single-port 800Gb adapters), each with a Gen5 x32 connection (two x16 connections)
- Up to 8x E3.S 1T NVMe drives for data storage, directly connected to a processor (4 to each processor), with a x4 host interface
 - PCIe adapters and drives are mutually exclusive; supported combinations are as follows:
 - 2x drives can be used with 5x adapters
 - 4x drives can be used with 4x adapters
 - 6x drives can be used with 3x adapters
 - 8x drives can be used with 2x adapters
- 2x E3.S 1T NVMe drives for OS boot, connected via a switched x4 host interface to CPU 2
- 2x 25Gb Ethernet SFP28 onboard connectors based on Mellanox ConnectX-6 Lx controller

AI Training configuration:

- 4x B200 GPU Modules connected to processors with x16 interface (2 to each processor)
- 4x ConnectX-8 I/O modules to enable high-speed PCIe Gen6 x16 GPUDirect connectivity to the GPUs and Gen5 x16 connectivity to the processors
- Each ConnectX-8 switch has directly connected to it:
 - 1x B200 GPU Module via Gen6 x16
 - 1x Grace processor via Gen5 x16
 - 1x 800 Gb OSFP network connection
 - 2x E3.S drives for storage
- Additional network connectivity options:
 - 2x ConnectX-7 (single-port 400Gb adapters or dual-port 200Gb adapters), each with a Gen5 x16 connection
 - 1x ConnectX-8 (single-port 800Gb adapter), with a Gen5 x32 connection (two x16 connections)
- 2x E3.S 1T NVMe drives for OS boot, connected via a switched x4 host interface to CPU 2
- 2x 25Gb Ethernet SFP28 onboard connectors based on Mellanox ConnectX-6 Lx controller

The feature codes listed in the following table form the building blocks used to create the above configurations.

Table 4. Configurations

Configuration	Feature code	Description	Qty
HPC and AI for Science configuration	C8X1	ThinkSystem SC777 V4 MCIO Breakout Board	2
AI Training configuration	C8WZ	ThinkSystem SC777 V4 NVIDIA Dual XDR IO Module	2

Configurator tips:

- The AI Training configuration is the default in DCSC, feature C8WZ
- To configure the server for HPC and AI for Science, do the following:
 1. In the **PCI > Network Module** section, change Network Module from C8WZ to None
 2. In the **PCI > InfiniBand Adapter** section, select up to 6 network adapters to be installed in the PCIe slots
 3. Feature C8X1 will automatically be derived

NVIDIA GB200 NVL4 HPM

The SC777 V4 leverages two NVIDIA GB200 NVL4 host processor motherboards (HPMs) per tray that integrate processors, memory and GPU accelerators in a powerful package.

- [Processors](#)
- [Memory](#)
- [GPU accelerators](#)

Processors

Each of the NVIDIA GB200 NVL4 HPMs include one NVIDIA Grace processor, as part of the NVIDIA GB200 NVL4 module. The processors are standard in all configurations of the SC777 V4.

Each processor has the following features:

- 72 Arm Neoverse V2 cores with 4x 128b SVE2
- 3.1 GHz base frequency
- 3.0 GHz all-core SIMD frequency
- L1 cache: 64KB i-cache + 64KB d-cache
- L2 cache: 1MB per core
- L3 cache: 114MB
- 64 PCIe Gen5 lanes (implemented as 8x x8)
- 300 GB/s bi-directional NVLink-C2C connectivity between processors
- 450 GB/s bi-directional NVLink connectivity to two GPUs each
- NVIDIA Scalable Coherency Fabric (SCF) with 3.2 terabytes/s bisection bandwidth interconnecting Cores, Memory and IO
- 300W TDP (includes power for memory)

The NVIDIA Grace processor is integrated using Ball Grid Array (BGA) technology and is soldered onto the NVIDIA GB200 NVL4 HPM board. To replace the CPU, it is necessary to replace the entire module.

Memory

The NVIDIA GB200 NVL4 HPM boards incorporate processor memory via LPDDR5x modules.

Features of the memory subsystem are as follows:

- 960GB of LPDDR5X ECC memory in a fixed configuration, 480GB per processor
- Implemented using 8x 128GB modules (4 per CPU)
- Peak memory bandwidth per CPU: Up to 384 GB/s
- Memory clock per CPU: 4266MT/s

The Single Error Correction (SEC) and Double Error Detection (DED) algorithms as part of the ECC are designed to mitigate multi-bit aliasing and correct errors effectively. Furthermore, features such as parity, scrubbing, poisoning, and page off-lining are implemented to enhance Reliability, Availability, and Serviceability (RAS).

GPU accelerators

The NVIDIA GB200 NVL4 HPM boards support four onboard NVIDIA B200 GPUs. The following table lists the specifications for each GPU and for the full package. Measured performance will vary by application.

The following table lists the specifications for each GPU.

Table 5. GPU specifications

Specification	NVIDIA B200 GPU	GB200 NVL4 system
FP64	40 teraFLOPS	160 teraFLOPS
FP64 Tensor Core	40 teraFLOPS	160 teraFLOPS
FP32	80 teraFLOPS	320 teraFLOPS
TF32 Tensor Core	2,500 teraFLOPS*	10,000 teraFLOPS*
BFLOAT16 Tensor	5,000 teraFLOPS*	20,000 teraFLOPS*
FP16 Tensor Core	5,000 teraFLOPS*	20,000 teraFLOPS*
FP8 Tensor Core	10,000 teraFLOPS*	40,000 teraFLOPS*
FP4 Tensor Core	20,000 teraFLOPS*	80,000 teraFLOPS*
INT8 Tensor Core	10,000 TOPS*	40,000 TOPS*
GPU Memory	186 GB HBM3e per GPU	744 GB HBM3e
GPU Memory Bandwidth	8 TB/s	32 TB/s
GPU Memory ECC	Supported (enabled by default)	
Multi-Instance GPUs (MIG)	Up to 7 Instances	Up to 28 instances
Total Graphics Power (TGP)	1200W	5480W (total board)
Interconnect	NVLink between GPUs: 1.8 TB/s (full mesh) NVLink to CPUs: 450GB/s per GPU PCIe Gen6 x16: 256GB/s per GPU	

* With structural sparsity enabled

The following figure shows how the GPUs interconnect.

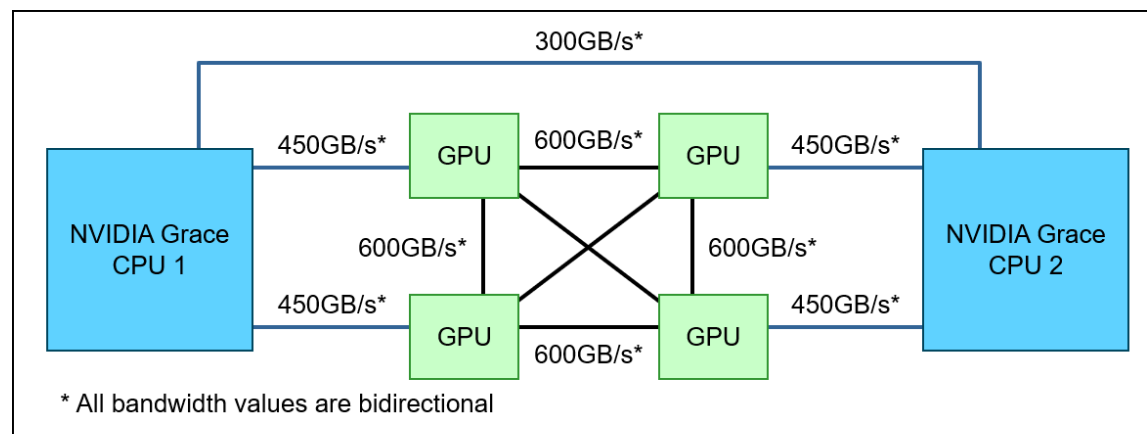


Figure 8. GB200-NVL4 NVLink topology

Internal storage

The SC777 V4 node supports up to 10x E3.S 1T SSDs. These are internal drives - they are not front accessible and are non-hot-swap.

- 8x drives for data storage
- 2x drives for OS boot, optional; feature C51T (ThinkSystem SC777 V4 Neptune PCIe Switch Board for Boot Drives) required

The drives are located as shown in the figure below.

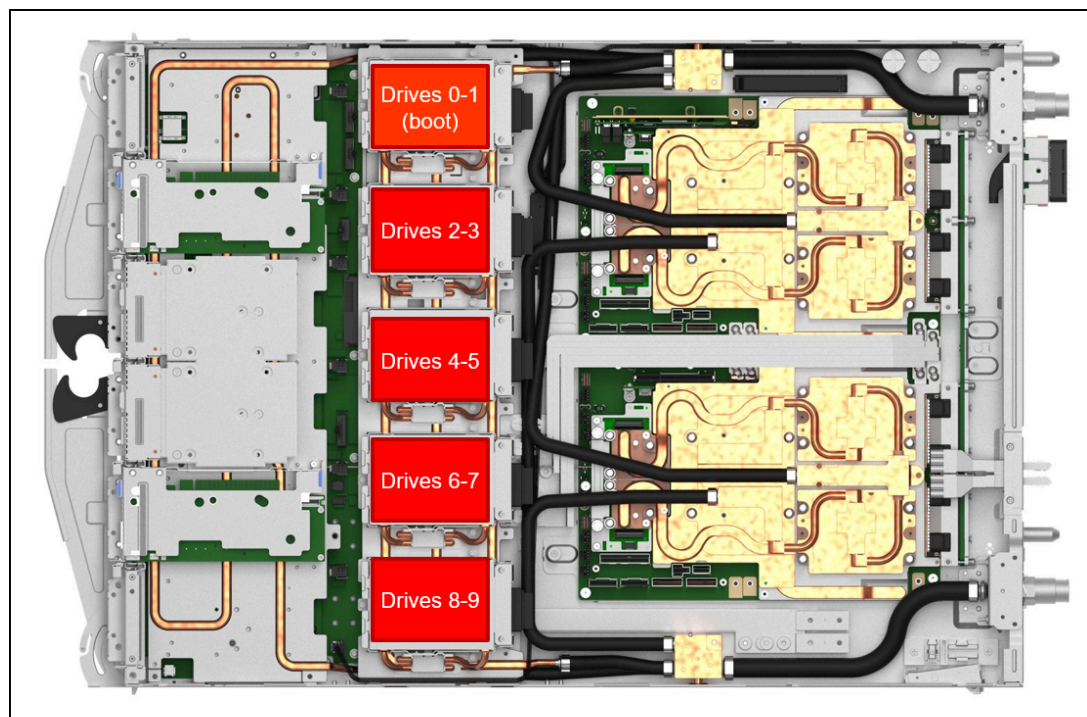


Figure 9. SC777 V4 internal drive bays

Each of the 5 drive bays supports either of the following:

- 2x E3.S 1T drives
- 1x E3.S 2T drive

Configuration notes:

- The node supports only supports E3.S NVMe drives
- The drives are connected to onboard controllers; RAID functionality is provided using the operating system features
- Boot drives are the same as other drives but use switched PCIe and may have lower performance. The boot drive kit (feature C51T) is only required when a total of 10 drives are installed. The boot drive kit is not field upgradable.
- In the HPC and AI for Science configuration (see the [Configurations](#) section), the same PCIe lanes can be used either for PCIe adapters or for NVMe drives. The following table lists the supported combinations.

Table 6. Drive and adapter combinations (HPC and AI for Science configuration)

Data drives (E3.S 1T)	Boot drives (E3.S 1T)	Front adapter slots
0 data drives	2 boot drives	6 adapters
2 data drives	2 boot drives	5 adapters
4 data drives	2 boot drives	4 adapters
6 data drives	2 boot drives	3 adapters
8 data drives	2 boot drives	2 adapters

- In the AI Training configuration (see the [Configurations](#) section), there is no restriction regarding the numbers of drives
- NVMe drives are connected to CPUs as follows:
 - Drives 0-1 are boot drives connected to CPU 2
 - Drives 2-5 are data storage drives connected to CPU 2
 - Drives 6-9 are data storage drives connected to CPU 1

The feature codes to select the appropriate storage cage are listed in the following table.

Tip: To configure boot drives, specify feature C51T plus either feature C77X (1x boot drive) or C51U (2x boot drives). See the table below. The boot drive kit is only required when 8 performance and 2 boot drives are installed.

Table 7. Drive mounting kits

Feature code	Description	Max qty	Purpose
C51T	ThinkSystem SC777 V4 Neptune PCIe Switch Board for Boot Drives	1	Enables the use of boot drives
C77X	ThinkSystem SC777 V4 Neptune Single E3.S 1T Kit	3*	Cage for a single drive
C51U	ThinkSystem SC777 V4 Neptune Dual E3.S 1T Kit	5*	Cage for two drives (not for E3.S 2T drives)

* See the table below for guidance on quantities required

Each drive bay supports 1x or 2x E3.S 1T drives. If a bay is to have only 1 drive, order feature C77X for that bay; if a bay is to have 2 drives, order feature C51U for that bay. The following table lists the required quantities of each drive bay cage

Table 8. Quantities of drive cages required based on desired number of drives

Desired drive count	Quantity of single-drive cages required (feature C77X)	Quantity of dual-drive cages required (feature C51U)
Boot drives (bays 0-1)		
1 drive	1x C77X	0
2 drives	0	1x C51U
Data storage drives (bays 2-9)		
1 drive	1x C77X	0
2 drives	2x C77X	0
2 drives	0	1x C51U
3 drives	1x C77X	1x C51U
4 drives	0	2x C51U
5 drives	1x C77X	2x C51U

Desired drive count	Quantity of single-drive cages required (feature C77X)	Quantity of dual-drive cages required (feature C51U)
6 drives	0	3x C51U
7 drives	1x C77X	3x C51U
8 drives	0	4x C51U

Controllers for internal storage

The drives of the SC777 V4 are connected to integrated NVMe storage controllers.

If desired, RAID functionality is provided by the installed operating system (software RAID)

Internal drive options

The following tables list the drive options for internal storage of the server.

Trayless drives:

- [E3.S EDSFF trayless PCIe 5.0 NVMe SSDs](#)

SED support: The tables include a column to indicate which drives support SED encryption. The encryption functionality can be disabled if needed. Note: Not all SED-enabled drives have "SED" in the description.

Table 9. E3.S EDSFF trayless PCIe 5.0 NVMe SSDs

Part number	Feature code	Description	SED support	Max Qty
E3.S trayless SSDs - PCIe 5.0 NVMe - Mixed Use/Mainstream (3-5 DWPD)				
4XB7B12184	CEC4	ThinkSystem E3.S 1T VA 1.6TB Mixed Use NVMe PCIe 5.0 x4 Trayless SSD	Support	10
4XB7B12185	CEC5	ThinkSystem E3.S 1T VA 3.2TB Mixed Use NVMe PCIe 5.0 x4 Trayless SSD	Support	10
4XB7B12186	CEC6	ThinkSystem E3.S 1T VA 6.4TB Mixed Use NVMe PCIe 5.0 x4 Trayless SSD	Support	10
4XB7B12187	CEC7	ThinkSystem E3.S 1T VA 12.8TB Mixed Use NVMe PCIe 5.0 x4 Trayless SSD	Support	10
E3.S trayless SSDs - PCIe 5.0 NVMe - Read Intensive/Entry (<3 DWPD)				
4XB7B12180	CE88	ThinkSystem E3.S 1T VA 1.92TB Read Intensive NVMe PCIe 5.0 x4 Trayless SSD	Support	10
4XB7B12181	CE87	ThinkSystem E3.S 1T VA 3.84TB Read Intensive NVMe PCIe 5.0 x4 Trayless SSD	Support	10
4XB7B12182	CE86	ThinkSystem E3.S 1T VA 7.68TB Read Intensive NVMe PCIe 5.0 x4 Trayless SSD	Support	10
4XB7B12183	CE85	ThinkSystem E3.S 1T VA 15.36TB Read Intensive NVMe PCIe 5.0 x4 Trayless SSD	Support	10

I/O expansion

The SC777 V4 node supports up to 6x front-accessible slots, depending on the configuration:

- Slots 1, 2, 3: PCIe 5.0 x16 slots connected to CPU 1
- Slots 4, 5, 6: PCIe 5.0 x16 slots connected to CPU 2

The location of the slots is shown in the following figure.

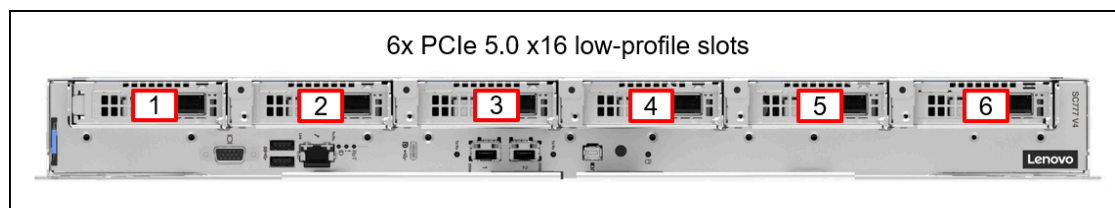


Figure 10. SC777 V4 PCIe slots

Slot availability is based on the configuration selected.

In the HPC and AI for Science configuration (see the [Configurations](#) section), PCIe adapters and data drives share the same PCIe connections to the processors. The following table lists the supported combinations of adapters and drives.

Table 10. Drive and adapter combinations (HPC and AI for Science configuration)

Front adapter slots	Data drives (E3.S 1T)	Boot drives (E3.S 1T)
1-2 adapters	8 data drives	2 boot drives
3 adapters	6 data drives	2 boot drives
4 adapters	4 data drives	2 boot drives
5 adapters	2 data drives	2 boot drives
6 adapters	0 data drives	2 boot drives

The AI Training configuration (see the [Configurations](#) section) supports:

- 2x OSFP interfaces installed in slots 3 and 4 (feature C8X0) (see [OSFP 800Gb ports](#) section)
- 2x PCIe adapters in slots 2 and 5

Slots are implemented using 1-slot risers. Ordering information for the risers plus the OSFP interface cage for AI Training configurations is in the following table.

Table 11. Slot ordering information

Feature code	Description	Slots where used	Max qty
HPC and AI for Science configuration			
C51P	ThinkSystem SC777 V4 Neptune Riser Kit - Slot 1	1, 6	2
C51N	ThinkSystem SC777 V4 Neptune Riser Kit - Slot 2/3/4/5/6	2, 3, 4, 5	4
AI Training configuration			
C8X0	ThinkSystem SC777 V4 OSFP IO Kit (two OSFP 800Gb cages)	3, 4	2
C51N	ThinkSystem SC777 V4 Neptune Riser Kit - Slot 2/3/4/5/6	2, 5	2

Onboard ports

The SC777 V4 has 2x 25GbE onboard ports. The ports are connected to an onboard NVIDIA ConnectX-6 Lx controller, implemented with SFP28 cages for optical or copper connections. Supports 1Gb, 10Gb and 25Gb connections. Locations of these ports is shown in the [Components and connectors](#) section.

25GbE Port 1 supports NC-SI for remote management. For factory orders, to specify which ports should have NC-SI enabled, use the feature codes listed in the [Remote Management](#) section. If neither is chosen, both ports will have NC-SI disabled by default.

For the specifications of the 25GbE ports including the supported transceivers and cables, see the ConnectX-6 Lx product guide:

<https://lenovopress.lenovo.com/lp1364-thinksystem-mellanox-connectx-6-lx-25gbe-sfp28-ethernet-adapters>

OSFP 800Gb ports

The AI Training configuration includes two OSFP interface boards (feature C8X0), each providing two OSFP ports. Supported connections are each up to 800 Gb/s (a total of 3200 Gb/s with 4x ports).

The following table lists the supported optical cables.

Table 12. Optical cables

Part number	Feature code	Description	NVIDIA part number
NVIDIA NDR Multi Mode Fibre Optical Cables			
4X97A81748	BQJN	Lenovo 3m NVIDIA NDR Multi Mode Optical Cable	980-9I73U-000003
4X97A81749	BQJP	Lenovo 5m NVIDIA NDR Multi Mode Optical Cable	980-9I73V-000005
4X97A81750	BQJQ	Lenovo 7m NVIDIA NDR Multi Mode MPO12 APC Optical Cable	980-9I57W-000007
4X97A81751	BQJR	Lenovo 10m NVIDIA NDR Multi Mode Optical Cable	980-9I57X-00N010
4X97A81752	BQJS	Lenovo 20m NVIDIA NDR Multi Mode Optical Cable	980-9I57Z-000020
4X97A85349	BSN6	Lenovo 30M NVIDIA NDR Multi Mode MPO12 APC Optical Cable	980-9I570-00N030
NVIDIA XDR SM MPO12 APC Optical Cable			
4X97B11375	CCVD	Lenovo 3M NVIDIA SM MPO12 APC Optical Cable (Generic)	980-9I55A-00N003
4X97B05747	C98S	Lenovo 5M NVIDIA XDR SM MPO12 APC Optical Cable (Generic)	980-9I55B-00N005
4X97B05748	C98T	Lenovo 10M NVIDIA XDR SM MPO12 APC Optical Cable (Generic)	980-9I58D-00N010
4X97B05750	C98U	Lenovo 15M NVIDIA XDR SM MPO12 APC Optical Cable (Generic)	980-9I58E-00N015
4X97B05751	C98V	Lenovo 20M NVIDIA XDR SM MPO12 APC Optical Cable (Generic)	980-9I58F-00N020
4X97B05752	C98W	Lenovo 30M NVIDIA XDR SM MPO12 APC Optical Cable (Generic)	980-9I58G-00N030
4X97B09056	CB2T	Lenovo 50M NVIDIA SM MPO12 APC Optical Cable (Generic)	980-9I58H-00N050
NVIDIA SM MPO12/APC to 2xSM MPO12/APC Split Optical Cable			
4X97B14779	CGUV	Lenovo 3M NVIDIA SM_MPO12/APC to 2xSM_MPO12/APC Optical Cable (Generic)	980-9I58K-00N003
4X97B14780	CGUW	Lenovo 5M NVIDIA SM_MPO12/APC to 2xSM_MPO12/APC Optical Cable (Generic)	980-9I58L-00N005
4X97B14781	CGUX	Lenovo 10M NVIDIA SM_MPO12/APC to 2xSM_MPO12/APC Optical Cable (Generic)	980-9I58N-00N010
4X97B14782	CGUY	Lenovo 15M NVIDIA SM_MPO12/APC to 2xSM_MPO12/APC Optical Cable (Generic)	980-9I56P-00N020
4X97B14783	CGUZ	Lenovo 20M NVIDIA SM_MPO12/APC to 2xSM_MPO12/APC Optical Cable (Generic)	980-9I56Q-00N030
4X97B14784	CGV0	Lenovo 30M NVIDIA SM_MPO12/APC to 2xSM_MPO12/APC Optical Cable (Generic)	980-9I56R-000050

The following table lists the supported direct-attach copper (DAC) cables.

Table 13. Copper cables

Part number	Feature code	Description	NVIDIA part number
NVIDIA NDRx2 OSFP800 Finned to NDRx2 OSFP800 Flat Copper Cables			
4X97A84581	BRKC	Lenovo 1M NVIDIA NDRx2 OSFP800 Finned to NDRx2 OSFP800 Flat Top Passive Copper Cable	980-9IA0H-00N001
4X97A84582	BRKD	Lenovo 1.5M NVIDIA NDRx2 OSFP800 Finned to NDRx2 OSFP800 Flat Top Passive Copper Cable	980-9IA0H-27N01A
4X97A84584	BRKF	Lenovo 3M NVIDIA NDRx2 OSFP800 Finned to NDRx2 OSFP800 Flat Top Active Copper Cable	980-9I601-00N003
NVIDIA NDRx2 OSFP800 to 2x NDR OSFP400 Splitter Copper Cables			
4X97A81827	BQJV	Lenovo 1m NVIDIA NDRx2 OSFP800 to 2x NDR OSFP400 Passive Copper Splitter Cable	980-9I432-00N001
4X97A81828	BQJW	Lenovo 1.5m NVIDIA NDRx2 OSFP800 to 2x NDR OSFP400 Passive Copper Splitter Cable	980-9I926-00N01A
4X97A81829	BQJX	Lenovo 2m NVIDIA NDRx2 OSFP800 to 2x NDR OSFP400 Passive Copper Splitter Cable	980-9I924-00N002
4X97A81830	BQJY	Lenovo 3m NVIDIA NDRx2 OSFP800 to 2x NDR OSFP400 Passive Copper Splitter Cable	980-9I92N-00N003
NVIDIA XDR OSFP to 2xOSFP Active Copper Splitter Cable			
4X97B05741	C98M	Lenovo 1m NVIDIA XDR OSFP to 2xOSFP Active Copper Splitter cable (Generic)	980-9IA05-00X001
4X97B05742	C98N	Lenovo 1.5m NVIDIA XDR OSFP to 2xOSFP Active Copper Splitter cable (Generic)	980-9IA05-00X01A
4X97B05743	C98P	Lenovo 2m NVIDIA XDR OSFP to 2xOSFP Active Copper Splitter cable (Generic)	980-9IA05-00X002
Lenovo IB Twin Port NDR OSFP112 to 2x NDR OSFP Splitter Copper Cables			
4X97B16876	CK2A	Lenovo 1M Luxshare 400Gx2 OSFP112 to 2x 400G OSFP Passive Copper Splitter Cable	-
4X97B16877	CK2B	Lenovo 1.5M Luxshare 400Gx2 OSFP112 to 2x 400G OSFP Passive Copper Splitter Cable	-
4X97B16878	CK2C	Lenovo 2M Luxshare 400Gx2 OSFP112 to 2x 400G OSFP Passive Copper Splitter Cable	-
4X97B16879	CK2D	Lenovo 3M Luxshare 400Gx2 OSFP112 to 2x 400G OSFP Passive Copper Splitter Cable	-

Network adapters

The SC777 V4 supports up to 6 network adapters installed in the front PCIe slots. The following table lists the supported adapters.

Table 14. Network adapters

Feature code	Description	Max qty	
		HPC and AI for Science configuration	AI Training configuration
BKSN	ThinkSystem NVIDIA ConnectX-7 NDR OSFP400 1-port PCIe Gen5 x16 InfiniBand Adapter DWC	6	2
BKSL	ThinkSystem NVIDIA ConnectX-7 NDR200/HDR/200GbE QSFP112 2-Port PCIe Gen5 x16 Adapter (SharedIO) DWC	6	2
CAGC	ThinkSystem NVIDIA ConnectX-8 8180 800Gbs XDR IB / 2x400GbE OSFP 1-Port PCIe Gen6 x16 DWC (Generic FW)	3	1

For more information, including the supported transceivers and cables, see the adapter product guides:

- ConnectX-7: <https://lenovopress.lenovo.com/lp1692-thinksystem-nvidia-connectx-7-ndr-infiniband-400gbe-osfp-adapters>
- ConnectX-8: <https://lenovopress.lenovo.com/lp2163-thinksystem-nvidia-connectx-8-8180-800gbs-osfp-pcie-gen6-x16-adapter>

Cooling

One of the most notable features of the ThinkSystem SC777 V4 offering is direct water cooling. Direct water cooling (DWC) is achieved by circulating the cooling water directly through cold plates that contact the CPU thermal case, DIMMs, drives, adapters, and other high-heat-producing components in the node.

One of the main advantages of direct water cooling is the water can be relatively warm and still be effective because water conducts heat much more effectively than air. Depending on the environmental conditions like water and air temperature, effectively 100% of the heat can be removed by water cooling; in configurations that stay slightly below that, the rest can be easily managed by a standard computer room air conditioner.

Allowable inlet temperatures for the water can be as high as 45°C (113°F) with the SC777 V4 for real-world applications. In most climates, water-side economizers can supply water at temperatures below 45°C for most of the year. This ability allows the data center chilled water system to be bypassed thus saving energy because the chiller is the most significant energy consumer in the data center. Typical economizer systems, such as dry-coolers, use only a fraction of the energy that is required by chillers, which produce 6-10°C (43-50°F) water. The facility energy savings are the largest component of the total energy savings that are realized when the SC777 V4 is deployed.

The advantages of the use of water cooling over air cooling result from water's higher specific heat capacity, density, and thermal conductivity. These features allow water to transmit heat over greater distances with much less volumetric flow and reduced temperature difference as compared to air.

For cooling IT equipment, this heat transfer capability is its primary advantage. Water has a tremendously increased ability to transport heat away from its source to a secondary cooling surface, which allows for large, more optimally designed radiators or heat exchangers rather than small, inefficient fins that are mounted on or near a heat source, such as a CPU.

The SC777 V4 offering uses the benefits of water by distributing it directly to the highest heat generating node subsystem components. By doing so, the offering realizes 7% - 10% direct energy savings when compared to an air-cooled equivalent. That energy savings results from the removal of the system fans and the lower operating temp of the direct water-cooled system components.

The direct energy savings at the enclosure level, combined with the potential for significant facility energy savings, makes the SC777 V4 an excellent choice for customers that are burdened by high energy costs or with a sustainability mandate.

Water is delivered to the enclosure from a coolant distribution unit (CDU) via water hoses, inlet and return. As shown in the following figure, each enclosure connects to the CDU via rack supply hoses.

The midplane in the enclosure routes the water via QuickConnects to each of the eight nodes, and each of the four power conversion stations.

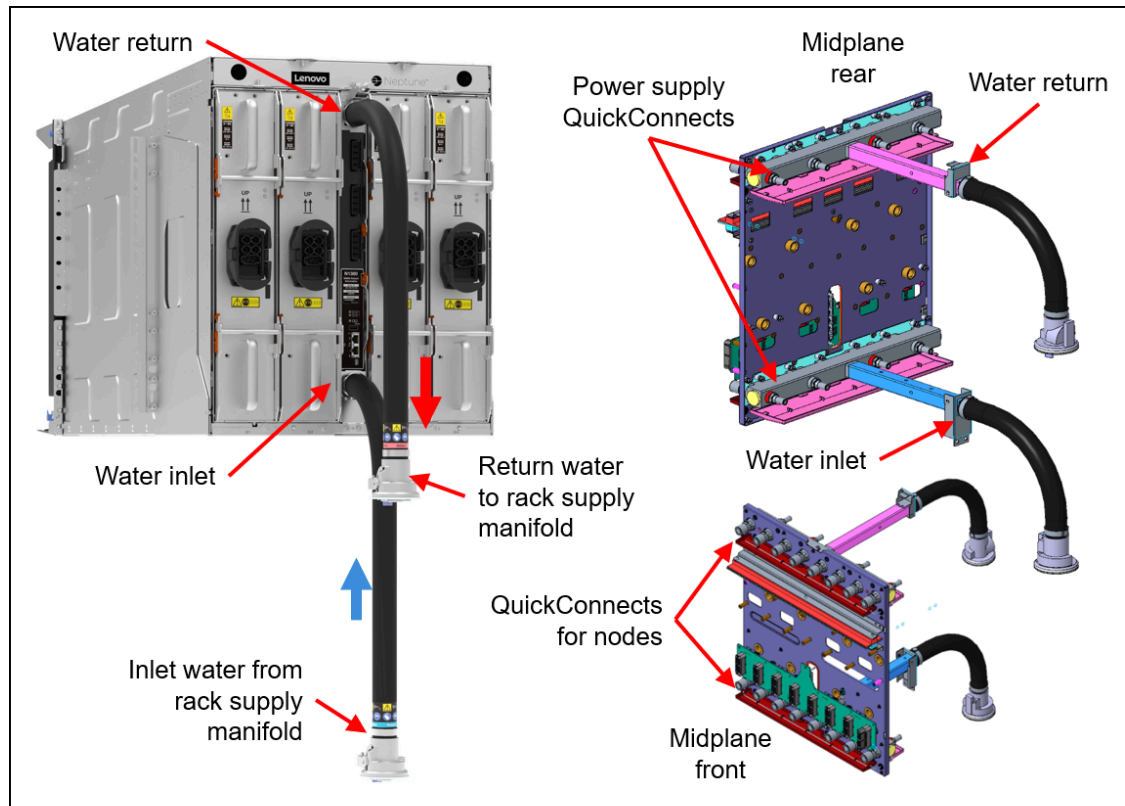


Figure 11. N1380 enclosure water connections

The water flows through the SC777 V4 tray to cool all major heat-producing components. The inlet water is split into two parallel paths, one for each node in the tray. Each path is then split further to cool the processors, memory, drives, and adapters.

During the manufacturing and test cycle, Lenovo's water-cooled nodes are pressure tested with Helium according to ASTM E499 / E499M – 11 (Standard Practice for Leaks Using the Mass Spectrometer Leak Detector in the Detector Probe Mode) and later again with Nitrogen to detect micro-leaks which may be undetectable by pressure testing with water and/or a water/glycol mixture as Helium and Nitrogen have smaller molecule sizes.

This approach also allows Lenovo to ship the systems pressurized without needing to send hazardous antifreeze-components to our customers.

Onsite the materials used within the water loop from the CDU to the nodes should be limited to copper alloys with brazed joints, Stainless steels with TIG and MIG welded joints and EPDM rubber. In some instances, PVC might be an acceptable choice within the facility.

The water the system is filled with must be reasonably clean, bacteria-free water (< 100 CFU/ml) such as de-mineralized water, reverse osmosis water, de-ionized water, or distilled water. It must be filtered with in-line 50 micron filter. Biocide and Corrosion inhibitors ensure a clean operation without microbiological growth or corrosion. For details about the implementation of Lenovo Neptune water cooling, see the document [Lenovo Neptune Direct Water-Cooling Standards](#).

Lenovo Data Center Power and Cooling Services can support you in the design, implementation and maintenance of the facility water-cooling infrastructure.

Water connections

Water connections to the N1380 enclosure are provided using hoses that connect directly to the coolant distribution unit (CDU), either an in-rack CDU or in-row CDU. As shown in the following figure, hose lengths required depend on the placement of the N1380 in the rack cabinet.

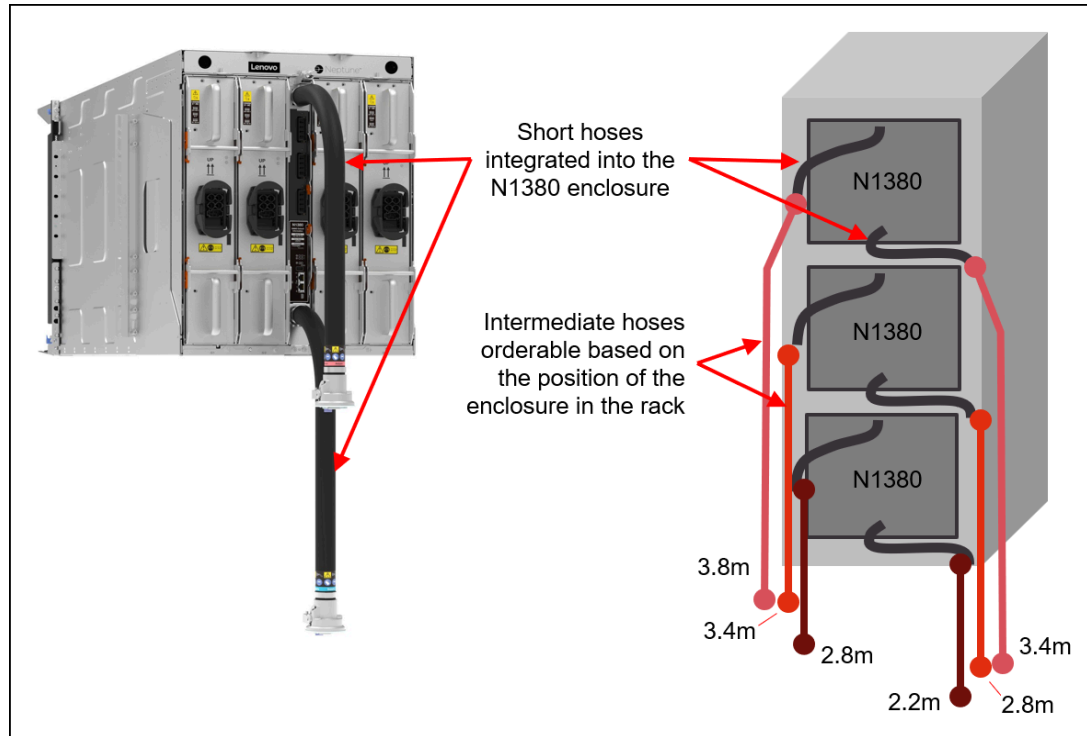


Figure 12. Hose lengths based on the position of the N1380 enclosure in the rack cabinet

Water connections are made up of the following:

- Short hoses permanently attached to the rear of the N1380 enclosure
- Intermediate hoses, orderable as three different pairs of hoses, based on the location of the N1380 enclosure in the rack. Ordering information for these hoses is listed in the table below.
- Connection to the CDU through additional hoses or valves on the data center water loop. Lenovo Datacenter Services can provide the end-to-end enablement for your water infrastructure needs.

The Intermediate hoses have a Sanitary Flange with a 1.5-inch Tri-Clamp at the CDU/data center end, as shown in the figure below.

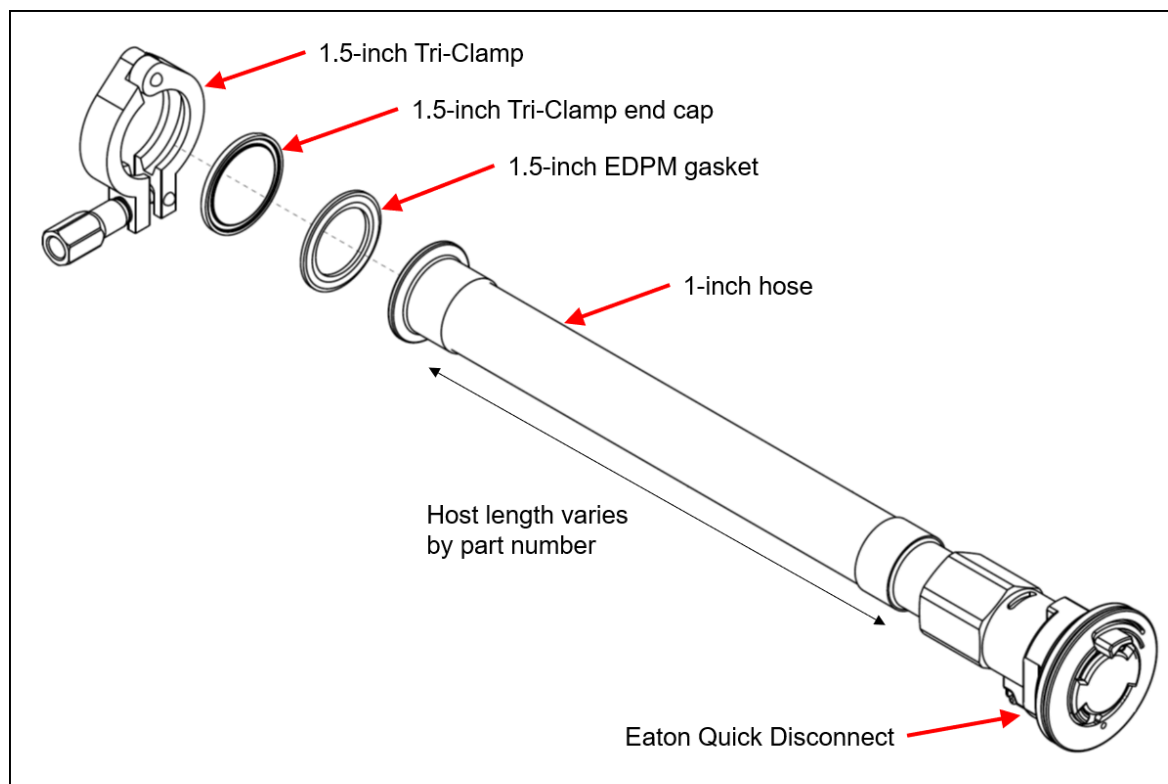


Figure 13. Connections of the Intermediate hoses

Ordering information for the Intermediate hoses to supply and return the water are listed in the following table. Hoses can either be stainless steel or EPDM, depending on the data center requirements.

Table 15. Water connection ordering information

Part number	Feature code	Description	Purpose
Stainless steel hoses			
CTO only	C2R0	ThinkSystem N1380 Neptune Stainless Steel Hose Connection	Internal water manifold for stainless steel hoses
4XH7A99480	C2KT	ThinkSystem N1380 Neptune 0.51M and 0.84M Stainless Steel Hose Set	Two stainless steel intermediate hoses for an N1380 installed starting in rack position U1-U9
4XH7A99481	C2KS	ThinkSystem N1380 Neptune 1.14M and 1.42M Stainless Steel Hose Set	Two stainless steel intermediate hoses for an N1380 installed starting in rack position U10-U22
4XH7A99482	C2KR	ThinkSystem N1380 Neptune 1.73M and 2M Stainless Steel Hose Set	Two stainless steel intermediate hoses for an N1380 installed starting in rack position U23-U36
EPDM hoses			
CTO only	BYKK	ThinkSystem N1380 Neptune EPDM Hose Connection	Internal water manifold for EPDM hoses
4XH7A99483	C2KQ	ThinkSystem N1380 Neptune 2.2M and 2.8M EPDM Hose Set	Two EPDM intermediate hoses for an N1380 installed starting in rack position U1-U9
4XH7A99484	C2KP	ThinkSystem N1380 Neptune 2.8M and 3.4M EPDM Hose Set	Two EPDM intermediate hoses for an N1380 installed starting in rack position U10-U22
4XH7A99485	C2KN	ThinkSystem N1380 Neptune 3.4M and 3.8M EPDM Hose Set	Two EPDM intermediate hoses for an N1380 installed starting in rack position U23-U36

For additional information, see the [Cooling](#) section.

To support the onsite setup for the direct water-cooled solution, a Commissioning Kit is available providing a flow meter, bleed hose, pressure gauge and vent valve. Ordering information is listed in the following table.

Table 16. Commissioning Kit field upgrade

Part number	Description
4XF7B07632	Neptune DWC Manifold Extended Commissioning Kit

Power conversion stations

The N1380 enclosure supports up to four water-cooled Power Conversion Stations. The use of water-cooled power conversion stations enables an even greater amount of heat can be removed from the data center using water instead of air-conditioning.

The Power Conversion Stations supply internal system power to a 48V busbar. This innovative design merges power conversion, rectification, and distribution into a single unit, a departure from traditional setups that require separate units, including separate power supplies, resulting in best-in-class efficiency.

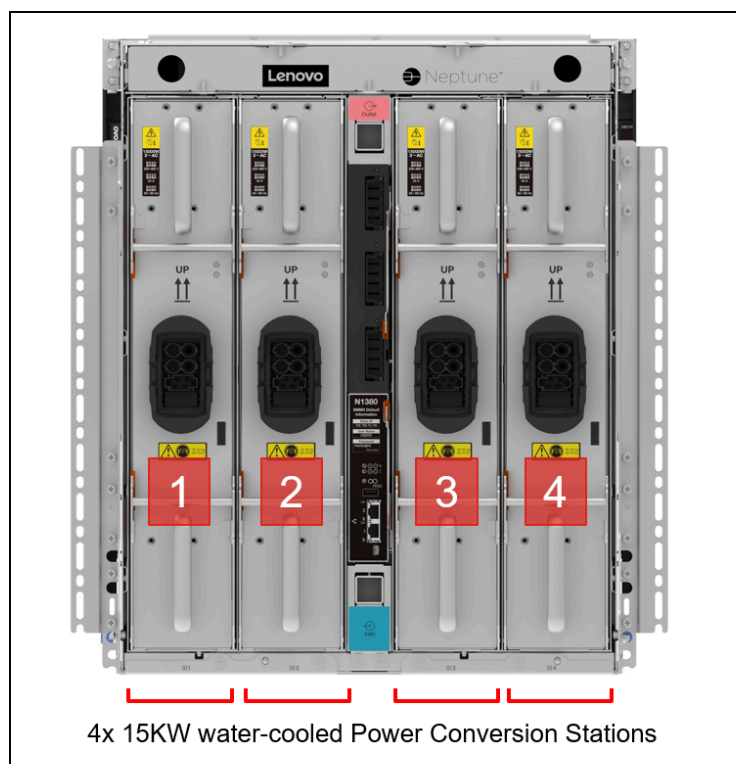


Figure 14. N1380 power conversion stations

The N1380 supports the power conversion stations listed in the following table.

Table 17. N1380 power conversion stations

Part number	Feature code	Description	Quantities	Efficiency	Connector	Input
4P57A82022	BYKH	ThinkSystem N1380 Neptune 15kW 3-Phase 380-480V Titanium Power Conversion Station	2, 3, 4	Titanium	Harting Han-ECO 10B	3-phase 380V-480Vac 30A per phase

Quantities of 2, 3 or 4 are supported. The power conversion stations provide N+1 redundancy with oversubscription.

Tip: Use Lenovo Capacity Planner to determine the power needs for your rack installation. See the [Lenovo Capacity Planner](#) section for details.

The power conversion stations have the following features:

- 80 PLUS Titanium certified
- Input: 380~480Vac, 3-phase WYE
- Input current: 30A per phase)
- N+1 power redundancy
- Power management configured through the SMM
- Built-in overload and surge protection
- Full telemetry of PCS available through SMM GUI or Redfish including: energy, power, voltage, current, and overall conversion efficiency

Power policies and power output

The following table lists the enclosure power capacity based on the power policy and the number of power modules installed.

N+0 and N+N not supported: The SC777 V4 does not support N+0 (non-redundant) or N+N power supplies in the N1380 enclosure.

Table 18. Power capacity based on power policy selected

Number of PCS units in Nominal State	N+1 redundancy	
	N+1 Policy (see notes)	Chassis capacity (DC)
1	-	-
2	1+1	18 KW (15 KW)*
3	2+1	36 KW (30 KW)*
4	3+1	54 KW (45 KW)*

* The numbers in parentheses are the chassis power capacity if no throttling is required when the redundancy is degraded (e.g. 3+1 and 1 PSU fails)

Notes:

- With N+1, if there is a PSU fault, the power capacity is reduced to <= the value shown in parentheses within 1 second. Example: If 3+1 policy is selected & all 4 PSUs are operational, the power capacity is 54KWDC. If a PSU faults, the power is reduced to <=45KW within 1 second. GPU trays are hard throttled w PWRBRK#. Intel trays are proportionally throttled with Psys power capping
- If the chassis is in the degraded redundancy state, due to 1 or more PSU faults, and an additional PSU fails, the chassis will attempt to stay alive by hard-throttling the trays (via PROCHOT# & PWRBRK#), but there is no guarantee. There is risk that the chassis could power off if the PSU rating is exceeded
- For the redundant power policies, only OVS modes are selectable in SMM (ie. OVS is always enabled). When a user creates a hardware config in DCSC, a chassis max power estimate is completed, via an LCP API call. If the chassis DC max power is in between the chassis degraded power capacity shown in parentheses and the chassis normal power capacity, a warning is displayed stating that the server could be throttled if a PSU fails. If the customer doesn't want potential throttling, they either need to reduce their config or select a different combo of PSUs/policy to achieve a higher power budget

Power cables

The power conversion stations function at a 32A current, but the power cables enable two PCS units to share a single 63A connection from the data center. Consequently, for racks fully equipped with 12 power conversion stations, only six power connections from the data center are needed.

The N1380 enclosure supports the following power cables.

Table 19. Power cables

Part number	Feature code	Description	Connector	Quantity required
Power cables routed from the floor				
4L67A90301	BYKG	3M, 32A 240-415V, 3-Phase WYE IEC Floor Rack Power Cable	32A IEC plug (3ph+G+N)	1 per PCS
4L67A90172	BYKV	2.8M, 63A 240-415V, 3-Phase WYE IEC to Y-Splitter Rack Power Cable	60A IEC plug (3ph+G+N)	1 of each per 2 PCS units
4L67A97427	C4KW	0.95M, 63A 240-415V, 3-Phase Y-Splitter Floor Power Cable	60A IEC plug (3ph+G+N), use with BYKV	
Power cables routed from above the rack cabinet				
4L67A90302	BYKE	3M, 32A 240-415V, 3-Phase WYE IEC Top Rack Power Cable	32A IEC plug (3ph+G+N)	1 per PCS
4L67A90172	BYKV	2.8M, 63A 240-415V, 3-Phase WYE IEC to Y-Splitter Rack Power Cable	60A IEC plug (3ph+G+N)	1 of each per 2 PCS units
4L67A90284	BYKF	0.95M, 63A 240-415V, 3-Phase Y-Splitter Top Power Cable	60A IEC plug (3ph+G+N), use with BYKV	

System Management

The SC777 V4 contains an integrated service processor, XClarity Controller 3 (XCC3), which provides advanced control, monitoring, and alerting functions. The XCC3 is based on the AST2600 baseboard management controller (BMC) using a dual-core ARM Cortex A7 32-bit RISC service processor running at 1.2 GHz.

Topics in this section:

- [System I/O Board \(DC-SCM\)](#)
- [Local management](#)
- [External Diagnostics Handset](#)
- [System status with XClarity One Mobile](#)
- [Remote management](#)
- [XCC3 Premier](#)
- [Remote management using the SMM3](#)
- [Lenovo XClarity Essentials](#)
- [Lenovo XClarity One](#)
- [Lenovo Capacity Planner](#)
- [Lenovo HPC & AI Software Stack](#)
- [NVIDIA AI Enterprise](#)

System I/O Board (DC-SCM)

The SC777 V4 implements a separate System I/O Board, also known as the DC-SCM (Data Center Secure Control Module, DCSCM), that connects to the system board as shown in the Internal view in the [Components and connectors](#) section. The System I/O Board contains connectors that are accessible from the exterior of the server as shown in the following figure.

Note: The NMI (non-maskable interrupt) button is no longer present on the board. Lenovo recommends using the NMI function that is part of the XCC user interfaces instead.

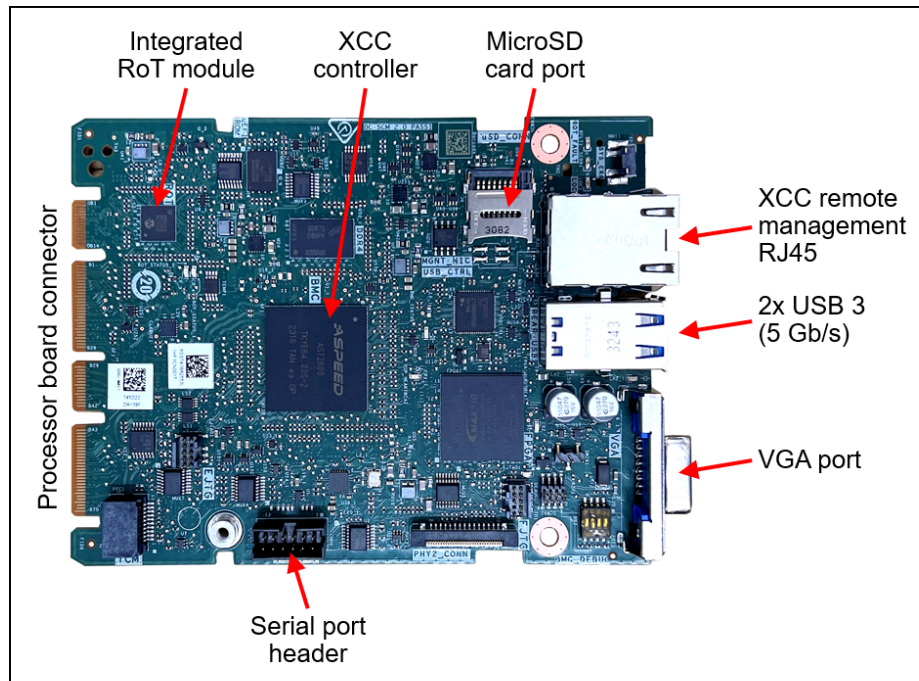


Figure 15. System I/O Board

The board also has the following components:

- XClarity Controller 3, implemented using the ASPEED AST2600 baseboard management controller (BMC).
- Root of Trust (RoT) module - implements Platform Firmware Resiliency (PFR) hardware Root of Trust (RoT) which enables the server to be NIST SP800-193 compliant. For more details about PFR, see the [Security](#) section.
- MicroSD card port to enable the use of a MicroSD card for additional storage for use with the XCC3 controller. XCC3 can use the storage as a Remote Disc on Card (RDOC) device (up to 4GB of storage). It can also be used to store firmware updates (including N-1 firmware history) for ease of deployment.

Tip: Without a MicroSD card installed, the XCC controller will have 100MB of available RDOC storage.

Ordering information for the supported Micro SD cards are listed in the [MicroSD for XCC local storage](#) section.

Local management

The following figure shows the ports and LEDs on the front of the SC777 V4.

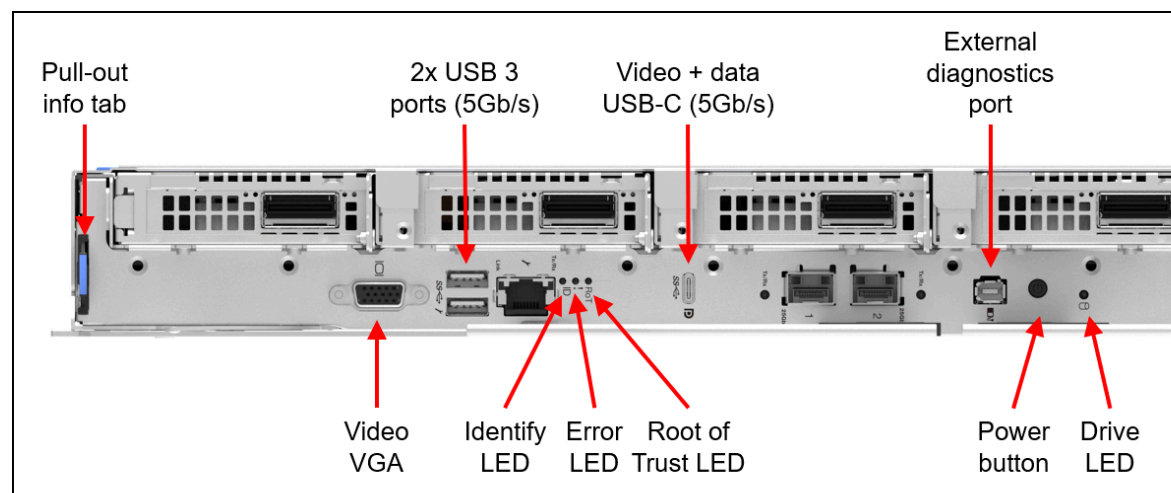


Figure 16. SC777 V4 Front operator panel

The LEDs are as follows:

- Identification (ID) LED (blue): Activated remotely via XCC to identify a specific server to local service engineers
- Error LED (yellow): Indicates if there is a system error. The error is reported in the XCC system log.
- Root of Trust fault LED (yellow): Indicates if there is an error with the Root of Trust (RoT) module
- Drive LED: Indicates activity on any drive

Information pull-out tab

The front of the server also houses an information pull-out tab (also known as the network access tag). See [Figure 2](#) for the location. A label on the tab shows the network information (MAC address and other data) to remotely access the service processor.

External Diagnostics Handset

The SC777 V4 has a port to connect an External Diagnostics Handset as shown in the following figure.

The External Diagnostics Handset allows quick access to system status, firmware, network, and health information. The LCD display on the panel and the function buttons give you access to the following information:

- Active alerts
- Status Dashboard
- System VPD: machine type & mode, serial number, UUID string
- System firmware levels: UEFI and XCC firmware
- XCC network information: hostname, MAC address, IP address, DNS addresses
- Environmental data: Ambient temperature, CPU temperature, GPU temperature
- Active XCC sessions
- System reset action

The handset has a magnet on the back of it to allow you to easily mount it on a convenient place on any rack cabinet.

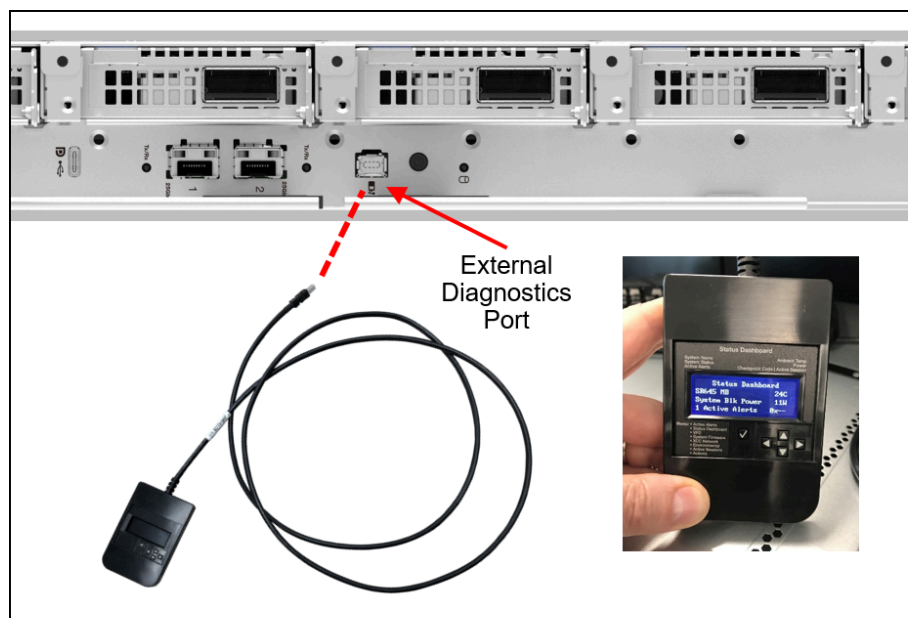


Figure 17. SC777 V4 External Diagnostics Handset

Ordering information for the External Diagnostics Handset with is listed in the following table.

Table 20. External Diagnostics Handset ordering information

Part number	Feature code	Description
4TA7A64874	1410 BEUX	ThinkSystem External Diagnostics Handset

System status with XClarity One Mobile

The XClarity One Mobile app includes a tethering function where you can connect your Android or iOS device to the server via USB to see the status of the server.

Download the XClarity One Mobile app from these sites:

- [Google Play Store](#)
- [Apple App Store](#)

The steps to connect the mobile device to the server are as follows:

1. In the XCC web interface under **BMC Configuration > Network**, set the USB mode to **Shared mode: Owned by BMC**
2. Connect the mobile device via a USB cable to the server's USB port with the management symbol 
3. In iOS or Android settings, enable Personal Hotspot or USB Tethering
4. Launch the Lenovo XClarity Mobile app

Once connected you can see the following information:

- Server status including error logs (read only, no login required)
- Server management functions (XClarity login credentials required)

Remote management

The 1Gb onboard port and one of the 25Gb onboard ports (port 1) on the front of the SC777 V4 offer a connection to the XCC for remote management. This shared-NIC functionality allows the ports to be used both for operating system networking and for remote management.

Remote server management is provided through industry-standard interfaces:

- Intelligent Platform Management Interface (IPMI) Version 2.0
- Simple Network Management Protocol (SNMP) Version 3 (no SET commands; no SNMP v1)
- Common Information Model (CIM-XML)
- Representational State Transfer (REST) support
- Redfish support (DMTF compliant)
- Web browser - HTML 5-based browser interface (Java and ActiveX not required) using a responsive design (content optimized for device being used - laptop, tablet, phone) with NLS support

The 25Gb Port 1 connector supports NC-SI. You can enable NC-SI in the factory using the feature code listed in the following table. If the feature code isn't selected, the port will have NC-SI disabled.

Table 21. Enabling NC-SI on the embedded network port

Feature code	Description
BEXY	ThinkSystem NC-SI enabled on SFP28 Port (Port 1)

IPMI via the Ethernet port (IPMI over LAN) is supported, however it is disabled by default. For CTO orders you can specify whether you want the feature enabled or disabled in the factory, using the feature codes listed in the following table.

Table 22. IPMI-over-LAN settings

Feature code	Description
B7XZ	Disable IPMI-over-LAN (default)
B7Y0	Enable IPMI-over-LAN

XCC3 Premier

In the SC777 V4, XCC3 has the Premier level of features built into the server. XCC3 Premier is equivalent to the XCC2 Premium offering in ThinkSystem V3 servers.

Configurator tip: Even though XCC3 Premier is a standard feature of the SC777 V4, it does *not* appear in the list of feature codes in the configurator.

XCC3 Premier includes the following functions:

- System Guard - Monitor hardware inventory for unexpected component changes, and simply log the event or prevent booting
- Neighbor Group - Enables administrators to manage and synchronize configurations and firmware level across multiple servers
- Syslog alerting
- Lenovo SED security key management
- Boot video capture and crash video capture
- Virtual console collaboration - Ability for up to 6 remote users to be log into the remote session simultaneously
- Remote console
- System utilization data and graphic view
- Single sign on with Lenovo XClarity Administrator
- Update firmware from a repository

- Enterprise Strict Security mode - Enforces CNSA 1.0 level security
- Remotely viewing video with graphics resolutions up to 1600x1200 at 75 Hz with up to 23 bits per pixel, regardless of the system state
- Remotely accessing the server using the keyboard and mouse from a remote client
- International keyboard mapping support
- Redirecting serial console via SSH
- Component replacement log (Maintenance History log)
- Access restriction (IP address blocking)
- Displaying graphics for real-time and historical power usage data and temperature
- Mapping the ISO and image files located on the local client as virtual drives for use by the server
- Mounting the remote ISO and image files via HTTPS, SFTP, CIFS, and NFS
- Power capping

With XCC3 Premier, for CTO orders, you can request that System Guard be enabled in the factory and the first configuration snapshot be recorded. To add this to an order, select feature code listed in the following table. The selection is made in the Security tab of the configurator.

Table 23. Enable System Guard in the factory (CTO orders)

Feature code	Description
BUT2	Enable System Guard

For more information about System Guard, see https://pubs.lenovo.com/xcc3/nn1ia_c_systemguard

Remote management using the SMM3

The N1380 enclosure includes a System Management Module 3 (SMM), installed in the rear of the enclosure. See [Enclosure rear view](#) for the location of the SMM. The SMM provides remote management of both the enclosure and the individual servers installed in the enclosure. The SMM can be accessed through a web browser interface and via Intelligent Platform Management Interface (IPMI) 2.0 commands.

The SMM provides the following functions:

- Remote connectivity to XCC controllers in each node in the enclosure
- Node-level reporting and control (for example, node virtual reseal/reset)
- Enclosure power management
- Enclosure thermal management
- Enclosure inventory

The following figure shows the LEDs and connectors of the SMM.

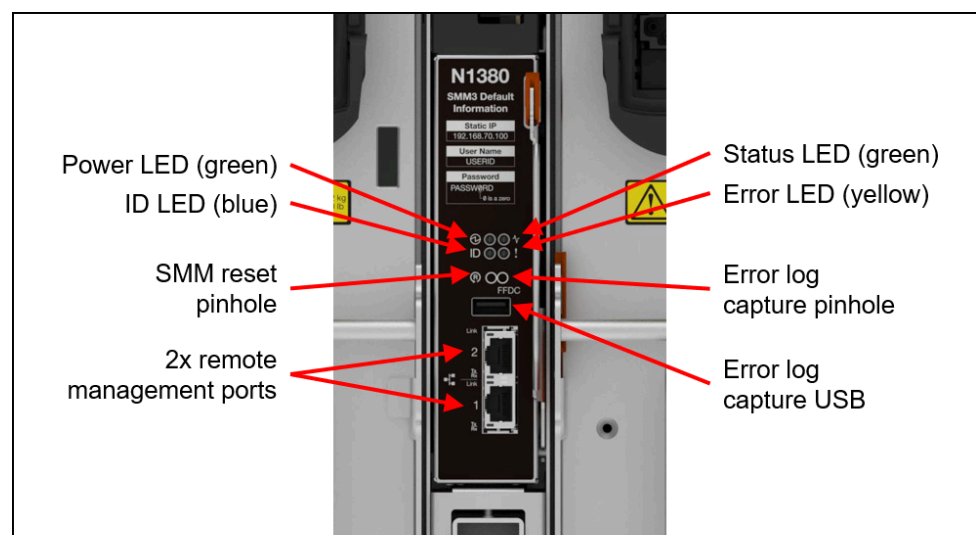


Figure 18. System management module in the N1380 enclosure

The SMM has the following ports and LEDs:

- 2x Gigabit Ethernet RJ45 ports for remote management access
- USB port and activation button for service
- SMM reset button
- System error LED (yellow)
- Identification (ID) LED (blue)
- Status LED (green)
- System power LED (green)

The USB service button and USB service port are used to gather service data in the event of an error. Pressing the service button copies First Failure Data Collection (FFDC) data to a USB key installed in the USB service port. The reset button is used to perform an SMM reset (short press) or to restore the SMM back to factory defaults (press for 4+ seconds).

The use of two RJ45 Ethernet ports enables the ability to daisy-chain the Ethernet management connections thereby reducing the number of ports you need in your management switches and reducing the overall cable density needed for systems management. With this feature you can connect the first SMM to your management network and the SMM in a second enclosure connects to the first SMM. The SMM in the third enclosure can then connect to the SMM in the second enclosure.

Up to 3 enclosures can be connected in a daisy-chain configuration and all 48 servers in those enclosures can be managed remotely via one single Ethernet connection.

Notes:

- If you are using IEEE 802.1D spanning tree protocol (STP) then at most 3 enclosures can be connected together
- Do not form a loop with the network cabling. The dual-port SMM at the end of the chain should not be connected back to the switch that is connected to the top of the SMM chain.

Lenovo XClarity Essentials

Lenovo offers the following XClarity Essentials software tools that can help you set up, use, and maintain the server at no additional cost:

- Lenovo Essentials OneCLI

OneCLI is a collection of server management tools that uses a command line interface program to manage firmware, hardware, and operating systems. It provides functions to collect full system health information (including health status), configure system settings, and update system firmware and drivers.

- Lenovo Essentials UpdateXpress

The UpdateXpress tool is a standalone GUI application for firmware and device driver updates that enables you to maintain your server firmware and device drivers up-to-date and help you avoid unnecessary server outages. The tool acquires and deploys individual updates and UpdateXpress System Packs (UXSPs) which are integration-tested bundles.

- Lenovo Essentials Bootable Media Creator

The Bootable Media Creator (BOMC) tool is used to create bootable media for offline firmware update.

ThinkSystem V4 servers: The format of UEFI and BMC settings has changed for ThinkSystem V4 servers, to align with OpenBMC and Redfish requirements. See the documentation of these tools for details. As a result, the following versions are required for these servers:

- OneCLI 5.x or later
- UpdateXpress 5.x or later
- BOMC 14.x or later

For more information and downloads, visit the Lenovo XClarity Essentials web page:

<http://support.lenovo.com/us/en/documents/LNVO-center>

Lenovo XClarity One

Lenovo XClarity One is a cloud-native unified Management-as-a-Service (MaaS) platform, engineered to modernize IT infrastructure. .

XClarity One is the next milestone in Lenovo's portfolio of systems management products. Now you can leverage the benefits of a true next-generation platform for the deployment, management, and maintenance of your infrastructure through a single, centralized platform that delivers a consistent user experience across all Lenovo products.

XClarity One's primary architectural value is divided into three key areas:

- How it predicts issues (AI/Analytics)
- How it securely connects (Secure Management Hubs),
- How it actually configures the hardware at scale (Template-Driven Lifecycle Control)

AI-powered Automation

XClarity One harnesses the power of AI and predictive analytics to enhance the performance and reliability of your infrastructure with proactive protection.

- **AI-Powered Predictive Failure Analytics** - XC1 utilizes machine learning infrastructure to look at live device telemetries. The platform securely collects memory-diagnostic logs and pushes them into an active MPFA AI model infrastructure to analyze health patterns and predict probable hardware degradation before it manifests as a hard failure.
- **AI-Powered Call-Home** - A Call-Home serviceable event opens a support ticket automatically, leveraging AI technology (in the case of MPFA) for problem determination and fast resolution automatically dispatching parts and services, reducing service costs and minimizing downtime.

Secure Management Hub

Lenovo's proprietary Management Hub is an on-premises virtual appliance that acts as the bridge between your infrastructure and the cloud.

- **On-Premises Security with Cloud Flexibility** - your infrastructure has no direct connection to the cloud, greatly reducing your attack surface from external threats while still having the deployment benefits, flexibility, and scalability of a cloud solution.
- **Authentication and Authorization** - built on a Zero Trust Architecture and requiring OTP Application authentication for all users to handle the support of all customers' servers and client devices. Role-based access controls help define and restrict permissions based on user roles.

Template-Driven Lifecycle Control

Lenovo's unified management model eliminates fragmented server administration, allowing you to deploy, track, and maintain your entire infrastructure fleet from a single cloud-native dashboard.

- **Automated Configuration and Compliance:** Leverage scalable, repeatable device templates to rapidly deploy firmware updates, operating systems, and core baseboard management controller (BMC/UEFI) simultaneously across multiple sites. XClarity One continuously analyzes hardware compliance, shifting out-of-compliance systems to a critical status until a template re-deployment aligns them back to corporate standards.
- **Global Asset Visibility and Fleet Telemetry:** Centralize real-time tracking of component health, warranty status, and comprehensive hardware inventories. The platform aggregates live usage data, including memory allocation, power consumption, and processor thermal trends, giving you total operational awareness whether a server resides in a centralized data center or a remote edge environment.

Centralized management via Lenovo XClarity One can be integrated into your environment as an optional component. License information for XClarity One is listed in the following table.

Table 24. XClarity One license information

Part number	Feature code	Description
Starter licenses (limited to 10 devices per organization)		
7S0X001EWW	SF8Y	XClarity One - Starter, Per Endpoint w/1 Yr SW S&S
7S0X001FWW	SF8Z	XClarity One - Starter, Per Endpoint w/2 Yr SW S&S
7S0X001GWW	SF90	XClarity One - Starter, Per Endpoint w/3 Yr SW S&S
7S0X001HWW	SF91	XClarity One - Starter, Per Endpoint w/4 Yr SW S&S
7S0X001JWW	SF92	XClarity One - Starter, Per Endpoint w/5 Yr SW S&S
Standard licenses		
7S0X000LWW	SCJC	XClarity One - Standard, Per Endpoint w/1 Yr SW S&S
7S0X001CWW	SENW	XClarity One - Standard, Per Endpoint w/2Yr SW S&S
7S0X000MWW	SCJD	XClarity One - Standard, Per Endpoint w/3 Yr SW S&S
7S0X001DWW	SENX	XClarity One - Standard, Per Endpoint w/4Yr SW S&S
7S0X000NWW	SCJE	XClarity One - Standard, Per Endpoint w/5 Yr SW S&S
7S0X000VWW	SFQ1	XClarity One - Standard, Per Endpoint w/6Yr SW S&S
7S0X000WWW	SFQ2	XClarity One - Standard, Per Endpoint w/7Yr SW S&S

Licensing notes:

- One license is needed per physical managed device.
- Starter license is functionally identical to Standard license but is limited to 10 devices per organization maximum.

For more information, see these resources:

- Lenovo XClarity One datasheet:
<https://lenovopress.lenovo.com/ds0188-lenovo-xclarity-one>
- Lenovo XClarity One product guide:
<https://lenovopress.lenovo.com/lp1992-lenovo-xclarity-one>
- Migrating to Lenovo XClarity One: An Overview:
<https://lenovopress.lenovo.com/lp2429-migrating-to-lenovo-xclarity-one-an-overview>

Lenovo Capacity Planner

Lenovo Capacity Planner is a power consumption evaluation tool that enhances data center planning by enabling IT administrators and pre-sales professionals to understand various power characteristics of racks, servers, and other devices. Capacity Planner can dynamically calculate the power consumption, current, British Thermal Unit (BTU), and volt-ampere (VA) rating at the rack level, improving the planning efficiency for large scale deployments.

For more information, refer to the Capacity Planner web page:

<http://datacentersupport.lenovo.com/us/en/solutions/lno-lcp>

Lenovo HPC & AI Software Stack

The Lenovo HPC & AI Software Stack combines open-source with proprietary best-of-breed Supercomputing software to provide the most consumable open-source HPC software stack embraced by all Lenovo HPC customers.

It provides a fully tested and supported, complete but customizable HPC software stack to enable the administrators and users in optimally and environmentally sustainable utilizing their Lenovo Supercomputers.

The Lenovo HPC & AI Software Stack is built on the most widely adopted and maintained HPC community software for orchestration and management. It integrates third party components especially around programming environments and performance optimization to complement and enhance the capabilities, creating the organic umbrella in software and service to add value for our customers.

The key open-source components of the software stack are as follows:

- **Confluent Management**

Confluent is Lenovo-developed open-source software designed to discover, provision, and manage HPC clusters and the nodes that comprise them. Confluent provides powerful tooling to deploy and update software and firmware to multiple nodes simultaneously, with simple and readable modern software syntax.

- **Slurm Orchestration**

Slurm is integrated as an open source, flexible, and modern choice to manage complex workloads for faster processing and optimal utilization of the large-scale and specialized high-performance and AI resource capabilities needed per workload provided by Lenovo systems. Lenovo provides support in partnership with NVIDIA.

- **Slinky**

Slinky Bridge is a software solution that enables HPC and AI convergence by allowing Slurm and Kubernetes workloads to share the same compute infrastructure. Acting as the source of truth for scheduling decisions, Slurm extends its scheduling, accounting, and resource management capabilities to Kubernetes workloads while maintaining support for traditional HPC jobs. This unified approach reduces GPU fragmentation, improves resource utilization, and allows organizations to support both HPC and cloud-native AI workflows on a common platform.

- **Energy Aware Runtime**

EAR is a powerful European open-source energy management suite supporting anything from monitoring over power capping to live-optimization during the application runtime. Lenovo collaborates with Barcelona Supercomputing Centre (BSC) and Energy Aware Solutions (EAS) on the continuous development and support and offers three versions with differentiating capabilities.

- **Open OnDemand**

Open OnDemand (OOD) is an open source, web-based access portal that provides a unified Graphical User Interface (GUI) for seamless access to HPC and AI clusters. It enables users to securely access compute resources, submit and monitor jobs, and launch rich interactive applications, such as Jupyter Notebooks, VS Code, RStudio, and TensorBoard, directly from a standard web browser. Through these interactive environments, OOD enables end-users to run complete AI and data-science workflows based on popular frameworks like TensorFlow, PyTorch, RAPIDS, and more, without needing to manage the underlying cluster complexity. Supporting multiple resource managers, including Slurm, PBS Pro, LSF, and Kubernetes, Open OnDemand makes it easy to consolidate HPC and AI workloads on a shared infrastructure, significantly improving usability, accessibility, and overall productivity while eliminating the need for command-line expertise.

For more information and ordering information, see the Lenovo HPC & AI Software Stack product guide:

<https://lenovopress.lenovo.com/lp1651-lenovo-hpc-ai-software-stack>

The key non-open-source components of the software stack for the SC777 V4 are:

- NVIDIA CUDA

NVIDIA CUDA is a parallel computing platform and programming model for general computing on graphical processing units (GPUs). With CUDA, developers are able to dramatically speed up computing applications by harnessing the power of GPUs. When using CUDA, developers program in popular languages such as C, C++, Fortran, Python and MATLAB and express parallelism through extensions in the form of a few basic keywords. For more information, see the [NVIDIA CUDA Zone](#).

- NVIDIA HPC Software Development Kit

The NVIDIA HPC SDK C, C++, and Fortran compilers support GPU acceleration of HPC modeling and simulation applications with standard C++ and Fortran, OpenACC directives, and CUDA. GPU-accelerated math libraries maximize performance on common HPC algorithms, and optimized communications libraries enable standards-based multi-GPU and scalable systems programming. Performance profiling and debugging tools simplify porting and optimization of HPC applications, and containerization tools enable easy deployment on-premises or in the cloud. For more information, see the [NVIDIA HPC SDK](#).

NVIDIA AI Enterprise

The SC777 V4 is designed for [NVIDIA AI Enterprise](#), which is a comprehensive suite of artificial intelligence and data analytics software designed for optimized development and deployment in enterprise settings.

NVIDIA AI Enterprise includes workload and infrastructure management software known as [Base Command Manager](#). This software provisions the AI environment, incorporating the components such as the Operating System, Kubernetes (K8S), [GPU Operator](#), and [Network Operator](#) to manage the AI workloads.

Additionally, NVIDIA AI Enterprise provides access to ready-to-use open-sourced containers and frameworks from NVIDIA like NVIDIA NeMo, NVIDIA RAPIDS, NVIDIA TAO Toolkit, NVIDIA TensorRT and NVIDIA Triton Inference Server.

- **NVIDIA NeMo** is an end-to-end framework for building, customizing, and deploying enterprise-grade generative AI models; NeMo lets organizations easily customize pretrained foundation models from NVIDIA and select community models for domain-specific use cases.
- **NVIDIA RAPIDS** is an open-source suite of GPU-accelerated data science and AI libraries with APIs that match the most popular open-source data tools. It accelerates performance by orders of magnitude at scale across data pipelines.
- **NVIDIA TAO Toolkit** simplifies model creation, training, and optimization with TensorFlow and PyTorch and it enables creating custom, production-ready AI models by fine-tuning NVIDIA pretrained models and large training datasets.
- **NVIDIA TensorRT**, an SDK for high-performance deep learning inference, includes a deep learning inference optimizer and runtime that delivers low latency and high throughput for inference applications. TensorRT is built on the NVIDIA CUDA parallel programming model and enables you to optimize inference using techniques such as quantization, layer and tensor fusion, kernel tuning, and others on NVIDIA GPUs. <https://developer.nvidia.com/tensorrt-getting-started>
- **NVIDIA TensorRT-LLM** is an open-source library that accelerates and optimizes inference performance of the latest large language models (LLMs). TensorRT-LLM wraps TensorRT's deep learning compiler and includes optimized kernels from FasterTransformer, pre- and post-processing, and multi-GPU and multi-node communication. <https://developer.nvidia.com/tensorrt>
- **NVIDIA Triton Inference Server** optimizes the deployment of AI models at scale and in production for both neural networks and tree-based models on GPUs.

It also provides full access to the [NVIDIA NGC](#) catalogue, a collection of tested enterprise software, services and tools supporting end-to-end AI and digital twin workflows and can be integrated with MLOps platforms such as ClearML, Domino Data Lab, Run:ai, UbiOps, and Weights & Biases.

Finally, NVIDIA AI Enterprise introduced [NVIDIA Inference Microservices](#) (NIM), a set of performance-optimized, portable microservices designed to accelerate and simplify the deployment of AI models. Those containerized GPU-accelerated pretrained, fine-tuned, and customized models are ideally suited to be self-hosted and deployed on the SC777 V4.

Security

Topics in this section:

- [Security features](#)
- [Platform Firmware Resiliency - Lenovo ThinkShield](#)

Security features

The server offers the following electronic security features:

- Support for Platform Firmware Resiliency (PFR) hardware Root of Trust (RoT) - see the [Platform Firmware Resiliency](#) section
- Firmware signature processes compliant with FIPS and NIST requirements
- System Guard (part of [XCC3 Premier](#)) - Proactive monitoring of hardware inventory for unexpected component changes
- Administrator and power-on password
- Integrated Trusted Platform Module (TPM) supporting TPM 2.0

The server is NIST SP 800-147B compliant.

The following table lists the security options for the SC777 V4.

Platform Firmware Resiliency - Lenovo ThinkShield

Lenovo's ThinkShield Security is a transparent and comprehensive approach to security that extends to all dimensions of our data center products: from development, to supply chain, and through the entire product lifecycle.

The ThinkSystem SC777 V4 includes Platform Firmware Resiliency (PFR) hardware Root of Trust (RoT) which enables the system to be NIST SP800-193 compliant. This offering further enhances key platform subsystem protections against unauthorized firmware updates and corruption, to restore firmware to an integral state, and to closely monitor firmware for possible compromise from cyber-attacks.

PFR operates upon the following server components:

- UEFI image – the low-level server firmware that connects the operating system to the server hardware
- XCC image – the management “engine” software that controls and reports on the server status separate from the server operating system
- FPGA image – the code that runs the server’s lowest level hardware controller on the motherboard

The Lenovo Platform Root of Trust Hardware performs the following three main functions:

- Detection – Measures the firmware and updates for authenticity
- Recovery – Recovers a corrupted image to a known-safe image
- Protection – Monitors the system to ensure the known-good firmware is not maliciously written

These enhanced protection capabilities are implemented using a dedicated, discrete security processor whose implementation has been rigorously validated by leading third-party security firms. Security evaluation results and design details are available for customer review – providing unprecedented transparency and assurance.

The SC777 V4 includes support for Secure Boot, a UEFI firmware security feature developed by the UEFI Consortium that ensures only immutable and signed software are loaded during the boot time. The use of Secure Boot helps prevent malicious code from being loaded and helps prevent attacks, such as the installation of rootkits. Lenovo offers the capability to enable secure boot in the factory, to ensure end-to-end protection. Alternatively, Secure Boot can be left disabled in the factory, allowing the customer to enable it themselves at a later point, if desired.

The following table lists the relevant feature code(s).

Table 25. Secure Boot options

Part number	Feature code	Description	Purpose
CTO only	BPKQ	TPM 2.0 with Secure Boot	Configure the system in the factory with Secure Boot enabled.
CTO only	BPKR	TPM 2.0	Configure the system without Secure Boot enabled. Customers can enable Secure Boot later if desired.

Tip: If Secure Boot is not enabled in the factory, it can be enabled later by the customer. However once Secure Boot is enabled, it cannot be disabled.

Operating system support

The SC777 V4 supports the following operating systems:

- Red Hat Enterprise Linux 9.6
- Red Hat Enterprise Linux 10.0
- Ubuntu 24.04 LTS 64-bit

See Operating System Interoperability Guide (OSIG) for the complete list of supported, certified, and tested operating systems, including version and point releases:

<https://lenovopress.lenovo.com/osig#servers=sc777-v4-7dmp>

Also review the latest LeSI Best Recipe to see the operating systems that are supported via Lenovo Scalable Infrastructure (LeSI):

<https://support.lenovo.com/us/en/solutions/HT505184#5>

Physical and electrical specifications

Eight SC777 V4 server trays are installed in the N1380 enclosure. Each SC777 V4 tray has the following dimensions:

- Width: 546 mm (21.5 inches)
- Height: 53 mm (2.1 inches)
- Depth: 760 mm (29.9 inches) (799 mm, including the water connections at the rear of the server)

The N1380 enclosure has the following overall physical dimensions, excluding components that extend outside the standard chassis, such as EIA flanges, front security bezel (if any), and power conversion station handles:

- Width: 540 mm (21.3 inches)
- Height: 572 mm (22.5 inches)
- Depth: 1302 mm (51.2 inches)

The following table lists the detailed dimensions. See the figure below for the definition of each dimension.

Table 26. Detailed dimensions

Dimension	Description
483 mm	X_a = Width, to the outsides of the front EIA flanges
448 mm	X_b = Width, to the rack rail mating surfaces
540 mm	X_c = Width, to the outer most chassis body feature
572 mm	Y_a = Height, from the bottom of chassis to the top of the chassis
1062 mm	Z_a = Depth, from the rack flange mating surface to the rearmost I/O port surface
1076 mm	Z_b = Depth, from the rack flange mating surface to the rearmost feature of the chassis body
1114 mm	Z_c = Depth, from the rack flange mating surface to the rearmost feature such as power supply handle
226 mm	Z_d = Depth, from the forwardmost feature on front of EIA flange to the rack flange mating surface
226 mm	Z_e = Depth, from the front of security bezel (if applicable) or forwardmost feature to the rack flange mating surface

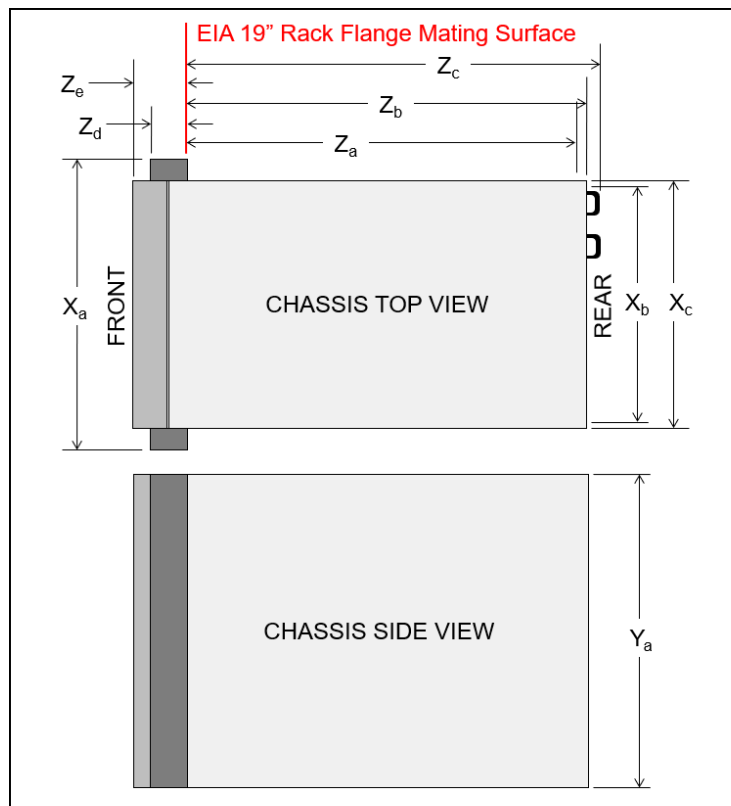


Figure 19. Enclosure dimensions

Shipping (cardboard packaging) dimensions are currently not available for the SC777 V4.

The shipping (cardboard packaging) dimensions of the N1380 are as follows:

- Width: 800 mm (31.5 inches)
- Height: 1027 mm (40.4 inches)
- Depth: 1600 mm (63.0 inches)

The SC777 V4 tray has the following maximum weight:

- 36.5 kg (80.5 lbs)

The N1380 enclosure has the following weight:

- Empty enclosure (with midplane and cables): 94 kg (208 lbs)
- Fully configured enclosure with 4x water-cooled power conversion stations and 8x SC777 V4 server trays: 470 kg (1036 lbs)

The enclosure has the following electrical specifications for input power conversion stations:

- Input voltage: 342~528Vac (nominal 380~480Vac), 3-phase WYE
- Input current (each power conversion station): 30A

Operating environment

The SC777 V4 server trays and N1380 enclosure are supported in the following environment.

Topics in this section:

- [Air temperature and humidity](#)
- [Water requirements](#)

- [Particulate contamination](#)

Air temperature and humidity

Air temperature/humidity requirements:

- Operating:
 - ASHRAE A3: 5°C to 40°C (41°F to 104°F); the maximum ambient temperature decreases by 1°C for every 175 m (574 ft) increase in altitude above 900 m (2,953 ft).
- Powered off: 5°C to 45°C (41°F to 113°F)
- Shipping/storage: -40°C to 60°C (-40°F to 140°F)
- Operating humidity:
 - ASHRAE Class A3: 8% to 85%; maximum dew point: 24°C (75°F)
- Shipment/storage humidity: 8% - 90%

Altitude:

- Maximum altitude: 3048 m (10 000 ft)

Water requirements

The water required to initially fill the system side cooling loop must be reasonably clean, bacteria-free water (<100 CFU/ml) such as de-mineralized water, reverse osmosis water, de-ionized water, or distilled water. The water must be filtered with an in-line 50 micron filter (approximately 288 mesh). The water must be treated with anti-biological and anti-corrosion measures. Environment quality must be maintain over the lifetime of the system to receive warranty and support on affecting components. For more information please see [Lenovo Neptune Direct Water-Cooling Standards](#).

In this section:

- [Pressure drop](#)
- [Water flow rates](#)

Pressure drop

See the following tables based on water temperature from the CDU.

Table 27. Pressure drop for S45 (45°C into the N1380 enclosure)

SC777 V4 quantity	1 power conversion station	2 power conversion stations	3 power conversion stations	4 power conversion stations
	Pressure drop (bar)	Pressure drop (bar)	Pressure drop (bar)	Pressure drop (bar)
1	0.63	0.64	0.65	0.65
2	0.64	0.65	0.66	0.67
3	0.66	0.67	0.69	0.70
4	0.68	0.70	0.72	0.73
5	0.71	0.73	0.75	0.77
6	0.75	0.77	0.79	0.83
7	0.79	0.82	0.84	0.88
8	0.84	0.87	0.89	0.93

Table 28. Pressure drop for S40 (40°C into the N1380 enclosure)

SC777 V4 quantity	1 power conversion station	2 power conversion stations	3 power conversion stations	4 power conversion stations
	Pressure drop (bar)	Pressure drop (bar)	Pressure drop (bar)	Pressure drop (bar)
1	0.46	0.46	0.46	0.48
2	0.47	0.47	0.48	0.50
3	0.48	0.49	0.50	0.52
4	0.50	0.51	0.52	0.54
5	0.53	0.54	0.55	0.57
6	0.55	0.57	0.58	0.62
7	0.58	0.60	0.62	0.65
8	0.62	0.64	0.66	0.70

Table 29. Pressure drop for S32 (32°C into the N1380 enclosure)

SC777 V4 quantity	1 power conversion station	2 power conversion stations	3 power conversion stations	4 power conversion stations
	Pressure drop (bar)	Pressure drop (bar)	Pressure drop (bar)	Pressure drop (bar)
1	0.29	0.29	0.30	0.31

SC777 V4 quantity	1 power conversion station	2 power conversion stations	3 power conversion stations	4 power conversion stations
2	0.30	0.30	0.31	0.32
3	0.31	0.32	0.32	0.34
4	0.32	0.33	0.34	0.35
5	0.34	0.35	0.35	0.37
6	0.36	0.37	0.38	0.39
7	0.38	0.39	0.40	0.42
8	0.40	0.41	0.43	0.45

Table 30. Pressure drop for S27 (27°C into the N1380 enclosure)

SC777 V4 quantity	1 power conversion station	2 power conversion stations	3 power conversion stations	4 power conversion stations
	Pressure drop (bar)	Pressure drop (bar)	Pressure drop (bar)	Pressure drop (bar)
1	0.25	0.25	0.25	0.25
2	0.25	0.26	0.26	0.26
3	0.26	0.27	0.27	0.28
4	0.27	0.28	0.28	0.30
5	0.29	0.29	0.30	0.32
6	0.30	0.31	0.32	0.33
7	0.32	0.33	0.34	0.35
8	0.34	0.35	0.36	0.38

Water flow rates

- 45°C water temperature
 - Water flow rate per SC777 V4 tray: 7.0 lpm
 - Water flow rate per PCS : 3.4 lpm
 - Water flow rate per enclosure: the sum of 7.0 lpm*(the number of trays) and 3.4 lpm*(the number of PCS)
- 40°C water temperature
 - Water flow rate per SC777 V4 tray: 6.0 lpm
 - Water flow rate per PCS : 2.8 lpm
 - Water flow rate per enclosure: the sum of 6.0 lpm*(the number of trays) and 2.8 lpm*(the number of PCS)
- 32°C water temperature
 - Water flow rate per SC777 V4 tray: 4.8 lpm
 - Water flow rate per PCS : 2.2 lpm
 - Water flow rate per enclosure: the sum of 4.8 lpm*(the number of trays) and 2.2 lpm*(the number of PCS)
- 27°C water temperature
 - Water flow rate per SC777 V4 tray: 4.4 lpm
 - Water flow rate per PCS : 2.0 lpm
 - Water flow rate per enclosure: the sum of 4.4 lpm*(the number of trays) and 2.0 lpm*(the number of PCS)

For example, when an N1380 enclosure is installed with eight SC777 V4 trays and four PCS, and the water temperature is 45°C, the enclosure water flow rate is 69.6 lpm (7.0 lpm*8 and 3.4 lpm*4)

For details on Lenovo Neptune direct water-cooling standards, see the following document:

Lenovo Neptune Direct Water-Cooling Standards

Particulate contamination

Airborne particulates (including metal flakes or particles) and reactive gases acting alone or in combination with other environmental factors such as humidity or temperature might damage the system that might cause the system to malfunction or stop working altogether.

The following specifications indicate the limits of particulates that the system can tolerate:

- Reactive gases:
 - The copper reactivity level shall be less than 200 Angstroms per month ($\text{\AA}/\text{month}$)
 - The silver reactivity level shall be less than 200 $\text{\AA}/\text{month}$
- Airborne particulates:
 - The room air should be continuously filtered with MERV 8 filters.
 - Air entering a data center should be filtered with MERV 11 or preferably MERV 13 filters.
 - The deliquescent relative humidity of the particulate contamination should be more than 60% RH
 - Environment must be free of zinc whiskers

For additional information, see the Specifications section of the documentation for the server, available from the Lenovo Documents site, <https://pubs.lenovo.com/>

Regulatory compliance

The SC777 V4 conforms to the following standards:

- ANSI/UL 62368-1
- IEC 62368-1 (CB Certificate and CB Test Report)
- CSA C22.2 No. 62368-1
- Mexico NOM-019
- South Africa NRCS LOA
- Ukraine UkrCEPRO
- Morocco CMIM Certification (CM)
- Russia, Belorussia and Kazakhstan, TP EAC 037/2016 (for RoHS)
- CE, UKCA Mark (EN55032 Class A, EN62368-1, EN55035, (EN61000-3-2, EN61000-3-3 or EN61000-3-11, EN61000-3-12) , (EU) 2019/424, and EN IEC 63000 (RoHS))
- FCC - Verified to comply with Part 15 of the FCC Rules, Class A
- Canada ICES-003, issue 7, Class A
- CISPR 32, Class A, CISPR 35
- Korea KS C 9832 Class A, KS C 9835
- Japan VCCI, Class A
- Taiwan BSMI CNS15936, Class A; Section 5 of CNS15663
- Australia/New Zealand AS/NZS CISPR 32, Class A; AS/NZS 62368.1
- SGS, VOC Emission
- Japanese Energy-Saving Act
- EU2019/424 Energy Related Product (ErP Lot9)

The N1380 conforms to the following standards:

- ANSI/UL 62368-1
- IEC 62368-1 (CB Certificate and CB Test Report)
- CSA C22.2 No. 62368-1
- Argentina IEC 62368-1
- Mexico NOM-019
- Ukraine UkrCEPRO
- Morocco CMIM Certification (CM)
- Russia, Belorussia and Kazakhstan, TP EAC 037/2016 (for RoHS)
- Russia, Belorussia and Kazakhstan, EAC: TP TC 004/2011 (for Safety); TP TC 020/2011 (for EMC)
- CE, UKCA Mark (EN55032 Class A, EN62368-1, EN55035, (EN61000-3-2, EN61000-3-3 or EN61000-3-11, EN61000-3-12) , (EU) 2019/424, and EN IEC 63000 (RoHS))
- FCC - Verified to comply with Part 15 of the FCC Rules, Class A
- Canada ICES-003, issue 7, Class A
- CISPR 32, Class A, CISPR 35
- Korea KS C 9832 Class A, KS C 9835
- Japan VCCI, Class A
- Australia/New Zealand AS/NZS CISPR 32, Class A; AS/NZS 62368.1
- SGS, VOC Emission
- China CELP certificate, HJ 2507-2011

Warranty upgrades and post-warranty support

The server and enclosure have the following warranty:

- Lenovo ThinkSystem SC777 V4 (7DMP) - 3-year warranty
- Lenovo ThinkSystem SC777 V4 (7DKA) - 5-year warranty
- Lenovo ThinkSystem N1380 Enclosure (7DDH) - 3-year warranty

Our global network of regional support centers offers consistent, local-language support enabling you to vary response times and level of service to match the criticality of your support needs:

- **Standard Next Business Day** – Best choice for non-essential systems requiring simple maintenance.
- **Premier Next Business Day** – Best choice for essential systems requiring technical expertise from senior-level Lenovo Engineers.
- **Premier 24x7 4-Hour Response** – Best choice for systems where maximum uptime is critical.
- **Premier Support Plus 24x7 4-Hour Response** – Best choice for systems where maximum uptime is critical.

For more information, consult our [Lenovo Operational Support Services for Data Centers Services Brochure](#).

Note: Some warranty upgrades may not be available in all markets or regions. For more information, go to <https://lenovolocator.com/>. For information about Lenovo warranty upgrade offerings that are available in your region, contact your local Lenovo sales representative or business partner.

GPU-related onsite support requests for the SC777 V4: Onsite Service Level targets (e.g. Next Business Day or 4 hours) normally apply after problem determination has been completed; however, these services levels **do not apply** for GPU-related service requests on the SC777 V4. Where a service agreement is in place, onsite service levels for GPU-related issues will be provided on a commercially reasonable endeavor's basis where applicable. Where customers require higher service levels for GPUs, purchasing onsite spare parts may improve target response times. For more information, see the [Lenovo ISG Services Agreement](#).

Services

Lenovo Infrastructure Services empower you at every stage of your IT lifecycle. From expert advisory and strategic planning to seamless deployment and ongoing support, we ensure your infrastructure is built for success. Our comprehensive services accelerate time to value, minimize downtime, and free your IT staff to focus on driving innovation and business growth.

Note: Some service options may not be available in all markets or regions. For more information, go to <https://lenovocator.com/>. For information about Lenovo service upgrade offerings that are available in your region, contact your local Lenovo sales representative or business partner.

In this section:

- [Lenovo Advisory Services](#)
- [Lenovo Implementation Services](#)
- [Infrastructure Deployment Services](#)
- [Lenovo Support Services](#)
- [Lenovo Managed Services](#)
- [Lenovo Sustainability Services](#)

Lenovo Advisory Services

Lenovo Advisory Services simplify the planning process, enabling customers to build future-proofed strategies in as little as six weeks. Consultants provide guidance on projects including VM migration, storage, backup and recovery, and cost management to accelerate time to value, improve cost efficiency, and build a flexibly scalable foundation.

- **Assessment Services**

An Assessment helps solve your IT challenges through an onsite, multi-day session with a Lenovo technology expert. We perform a tools-based assessment which provides a comprehensive and thorough review of a company's environment and technology systems. In addition to the technology-based functional requirements, the consultant also discusses and records the non-functional business requirements, challenges, and constraints. Assessments help organizations like yours, no matter how large or small, get a better return on your IT investment and overcome challenges in the ever-changing technology landscape.

Lenovo Implementation Services

Optimize your IT operations by shifting labor-intensive functions to Lenovo's skilled technicians for seamless on-site or remote deployment, configuration, and migration. Enjoy peace of mind, faster time to value, and comprehensive knowledge sharing with your IT staff, backed by our best-practice methodology.

Infrastructure Configuration Services

- **Factory Integrated Services**

A suite of value-added offerings provided during the manufacturing phase of a server or storage system that reduces time to value. These services aim at improving your hardware deployment experience and enhance the quality of a standard configuration before it arrives at your facility.

Infrastructure Deployment Services

Lenovo Infrastructure Deployment Services help you move from hardware delivery toward production with the planning, installation, configuration, integration, validation, documentation, and operational handoff support your project requires. Choose from clearly defined Premier Deploy for Infrastructure options, Specialized Deployment Services, or Hardware Installation Services based on your platform, project complexity, internal IT capabilities, and desired level of assurance.

For more information, consult our [Lenovo Infrastructure Deployment Services Catalog](#).

- **Premier Deploy for Infrastructure**

Clearly defined deployment options for eligible ThinkSystem servers, ThinkSystem storage arrays, and ThinkAgile solutions.

Table 31. Premier Deploy for Infrastructure

	Standard Deploy	Premier Deploy	Premier Deploy Plus
Choose when:	You need straightforward physical hardware installation and customer acceptance.	You need operating system installation and/or system configuration, integration and validation.	You need extended post-deployment assurance and visibility after implementation.
You gain:	An installed, accepted, and documented hardware foundation.	A validated, integrated environment ready to move into production.	Added assurance as your team transitions into ongoing operations.

For more information, see the [Premier Deploy for Infrastructure 1-pager](#).

- **Specialized Deployment Services**

Focused deployment expertise for specific technologies or project requirements when Premier Deploy for Infrastructure is not supported or is not the right fit for your project. Services can support ThinkSystem storage arrays, ThinkAgile solutions, HPC, liquid cooling, and private cloud environments through activities such as readiness planning, configuration, integration, validation, documentation, knowledge transfer, and operational handoff, as applicable. Availability and scope vary by platform, project requirements, geography, and service scope.

- **Hardware Installation Services**

Expert physical installation for eligible hardware outside the Premier Deploy for Infrastructure scope, or for projects that require installation only.

- **Hardware Installation Plus**

Extends Hardware Installation Services with:

- RAID configuration
- Supported operating system installation
- Driver-level verification

Lenovo Support Services

In addition to response time options for hardware parts, repairs, and labor, Lenovo offers a wide array of additional support services to ensure your business is positioned for success and longevity. Our goal is to reduce your capital outlays, mitigate your IT risks, and accelerate your time to productivity.

- **Premier Support for Infrastructure**

Helps you build business resilience with expert-led support, responsive issue resolution, and coordinated troubleshooting across supported Lenovo server and storage environments. Direct access to Lenovo expertise and accountable end-to-end case ownership help reduce disruption and simplify ongoing infrastructure operations.

- **Premier Support Plus for Infrastructure**

Gain all the benefits of Premier Support for Infrastructure, with proactive and predictive support capabilities, designated account oversight, preventative maintenance guidance, and Keep Your Drive to help identify potential issues earlier, plan service more effectively, and maintain control of sensitive data.

- **Committed Service Repair (CSR)**

Our commitment to ensuring the fastest, most seamless resolution times for mission-critical systems that require immediate attention to ensure minimal downtime and risk for your business. This service is only available for machines under the Premier 4-Hour Response SLA.

- **Multivendor Support Services (MVS)**

Your single point of accountability for resolution support across a vast range of leading Server, Storage, and Networking OEMs, allowing you to manage all your supported infrastructure devices seamlessly from a single source.

- **Keep Your Drive (KYD)**

Protect sensitive data and maintain compliance with corporate retention and disposal policies to ensure your data is always under your control, regardless of the number of drives that are installed in your Lenovo server.

- **Technical Account Manager (TAM)**

Your single point of contact to expedite service requests, provide status updates, and furnish reports to track incidents over time, ensuring smooth operations and optimized performance as your business grows.

- **Enterprise Software Support (ESS)**

Gain comprehensive, single-source, and global support for a wide range of server operating systems and Microsoft server applications.

For more information, consult our [Lenovo Operational Support Services for Data Centers Services Brochure](#).

Lenovo Managed Services

Achieve peak efficiency, high security, and minimal disruption with Lenovo's always-on Managed Services. Our real-time monitoring, 24x7 incident response, and problem resolution ensure your infrastructure operates seamlessly. With quarterly health checks for ongoing optimization and innovation, Lenovo's remote active monitoring boosts end-user experience and productivity by keeping your data center's hardware performing at its best.

- **Managed Services**

Lenovo Managed Services provides continuous 24x7 remote monitoring (plus 24x7 call center availability) and proactive management of your data center using state-of-the-art tools, systems, and practices by a team of highly skilled and experienced Lenovo services professionals.

Quarterly reviews check error logs, verify firmware & OS device driver levels, and software as needed. We'll also maintain records of the latest patches, critical updates, and firmware levels, to ensure your systems are providing business value through optimized performance.

Lenovo Sustainability Services

Lenovo Sustainability Services helps you address sustainability priorities across the IT lifecycle, from offsetting emissions associated with device purchases and extending equipment life to securely managing assets at end of life. Together, these services support more circular IT practices while helping you protect data, manage costs, and recover value from technology investments.

- **Asset Recovery Services**

Lenovo Asset Recovery Services (ARS) provides a secure, seamless solution for managing end-of-life IT assets, ensuring data is safely sanitized while contributing to a more circular IT lifecycle. By maximizing the reuse or responsible recycling of devices, ARS helps businesses meet sustainability goals while recovering potential value from their retired equipment. For more information, see the [Simplify IT Asset Recovery](#) page.

- **CO2 Offset Services**

Lenovo's CO2 Offset Services offer a simple and transparent way for businesses to take tangible action on their IT footprint. By integrating CO2 offsets directly into device purchases, customers can easily support verified climate projects and track their contributions, making meaningful progress toward their sustainability goals without added complexity.

- **Lenovo Certified Refurbished**

Lenovo Certified Refurbished offers a cost-effective way to support IT circularity without compromising on quality and performance. Each device undergoes rigorous testing and certification, ensuring reliable performance and extending its lifecycle. With Lenovo's trusted certification, you gain peace of mind while making a more sustainable IT choice.

- **Data Center Power and Cooling Services**

The Data Center Infrastructure team will provide solution design and implementation services to support the power and cooling needs of the multi-node chassis and multi-rack solutions. This includes designing for various levels of power redundancy and integration into the customer power infrastructure. The Infrastructure team will work with site engineers to design an effective cooling strategy based on facility constraints or customer goals and optimize a cooling solution to ensure high efficiency and availability. The Infrastructure team will provide the detailed solution design and complete integration of the cooling solution into the customer data center. In addition, the Infrastructure team will provide rack and chassis level commissioning and stand-up of the water-cooled solution which includes setting and tuning of the flow rates based on water temperature and heat recovery targets. Lastly, the Infrastructure team will provide cooling solution optimization and performance validation to ensure the highest overall operational efficiency of the solution.

Lenovo TruScale

Lenovo TruScale XaaS is your set of flexible IT services that makes everything easier. Streamline IT procurement, simplify infrastructure and device management, and pay only for what you use – so your business is free to grow and go anywhere.

Lenovo TruScale is the unified solution that gives you simplified access to:

- The industry's broadest portfolio – from pocket to cloud – all delivered as a service
- A single-contract framework for full visibility and accountability
- The global scale to rapidly and securely build teams from anywhere
- Flexible fixed and metered pay-as-you-go models with minimal upfront cost
- The growth-driving combination of hardware, software, infrastructure, and solutions – all from one single provider with one point of accountability.

For information about Lenovo TruScale offerings that are available in your region, contact your local Lenovo sales representative or business partner.

Rack cabinets

The N1380 enclosure is supported in the following racks:

- Lenovo EveryScale 42U Onyx Heavy Duty Rack Cabinet, model 1410-O42
- Lenovo EveryScale 42U Pearl Heavy Duty Rack Cabinet, model 1410-P42
- Lenovo EveryScale 48U Onyx Heavy Duty Rack Cabinet, model 1410-O48
- Lenovo EveryScale 48U Pearl Heavy Duty Rack Cabinet, model 1410-P48

Genie Material Lift

Considering the weight of the trays in the enclosure, an onsite material lift is required to allow service by a single person. If you do not already have a material lift available, Lenovo offers the Genie Lift GL-8 material lift as configurable option to the rack cabinets. Ordering information is listed in the following table.

Note: If neither the Genie Lift GL-8 nor the ServerLift SL-350x is available onsite when onsite service is required, the customer will be responsible for getting the system to a suitable work surface (with access to power) prior to service technician arrival and returning the system to the rack when service is complete prior to service technician departure.

Table 32. Genie Lift GL-8 ordering information

Part number	Description
4XF7B02087	Genie Lift GL-8 (Standard Base) Material Lift Option Kit • Genie GL-8 material lift • Load platform • Foot-release brake

Configuration rules:

- For the SC777 V4, you will also need to order the Genie Lift GL-8 Option Kit for N1380 and SC-Series Systems (4XF7B04136) which includes the components needed to safely remove and install a SC777 V4 tray or a Power Conversion Station if maintenance of these components is required.
- Optionally, Lenovo also offers a 4-Wheel Utility Cart (4XF7B05731).

Ordering information is listed in the following table.

Table 33. Ordering information for the Genie Lift fixtures and Utility Cart

Part number	Description
4XF7B04136	Genie Lift GL-8 Option Kit for N1380 and SC-Series Systems • Rotate fixture • Lift tool fixture
4XF7B05731	4-Wheel Utility Cart • Custom 4-wheel utility cart with a mechanism to store the load platform and lift tool fixture on the cart sides when not in use

Lenovo Financial Services

Why wait to obtain the technology you need now? No payments for 90 days and predictable, low monthly payments make it easy to budget for your Lenovo solution.

- **Flexible**

Our in-depth knowledge of the products, services and various market segments allows us to offer greater flexibility in structures, documentation and end of lease options.

- **100% Solution Financing**

Financing your entire solution including hardware, software, and services, ensures more predictability in your project planning with fixed, manageable payments and low monthly payments.

- **Device as a Service (DaaS)**

Leverage latest technology to advance your business. Customized solutions aligned to your needs. Flexibility to add equipment to support growth. Protect your technology with Lenovo's Premier Support service.

- **24/7 Asset management**

Manage your financed solutions with electronic access to your lease documents, payment histories, invoices and asset information.

- **Fair Market Value (FMV) and \$1 Purchase Option Leases**

Maximize your purchasing power with our lowest cost option. An FMV lease offers lower monthly payments than loans or lease-to-own financing. Think of an FMV lease as a rental. You have the flexibility at the end of the lease term to return the equipment, continue leasing it, or purchase it for the fair market value. In a \$1 Out Purchase Option lease, you own the equipment. It is a good option when you are confident you will use the equipment for an extended period beyond the finance term. Both lease types have merits depending on your needs. We can help you determine which option will best meet your technological and budgetary goals.

Ask your Lenovo Financial Services representative about this promotion and how to submit a credit application. For the majority of credit applicants, we have enough information to deliver an instant decision and send a notification within minutes.

Seller training courses

The following sales training courses are offered for employees and partners (login required). Courses are listed in date order.

1. VTT: HPC Product Update

2026-06-12 | 38 minutes | Employees Only

Join Teresa Chin and Sophia Li as they discuss updates to the HPC Supercomputing products.

Topics include:

ThinkSystem N1380 Neptune

ThinkSystem SC750 V4

ThinkSystem SC777 V4

Published: 2026-06-12

Length: 38 minutes

Start the training:

Employee link: Grow@Lenovo

Course code: DVHPC235

2. ThinkSystem Supercomputing Servers Powered by NVIDIA

2025-06-27 | 30 minutes | Employees and Partners

This course offers you information about the Lenovo SC777 V4 Neptune server, the first Lenovo server to use an Arm processor from NVIDIA. By the end of this course, you'll be able to list three features of the ThinkSystem SC777 V4 Neptune server, list three features of the ThinkSystem N1380 Neptune enclosure, describe two customer benefits of the ThinkSystem SC777 V4 Neptune server, and list four workload environments to which the SC777 V4 server is well suited.

Tags: DataCenter Products, NVIDIA, ThinkSystem

Published: 2025-06-27

Length: 30 minutes

Start the training:

Employee link: Grow@Lenovo

Partner link: [Lenovo 360 Learning Center](#)

Course code: SXXW2545

3. Q1 Solutions Launch Lenovo ThinkSystem SC777 V4 Neptune® Quick Hit

2025-04-15 | 5 minutes | Employees and Partners

Lenovo announces the Lenovo ThinkSystem SC777 V4 Neptune®, the first server in the supercomputing class powered by the NVIDIA GB200 platform. The Lenovo ThinkSystem SC777 V4 Neptune excels in HPC environments for intensive modeling and simulation workloads and Hybrid AI. It's designed to handle technical computing, grid deployments, and analytics workloads in various fields such as research, life sciences, energy, engineering, and financial simulation.

Tags: Artificial Intelligence (AI), High-Performance Computing (HPC), Nvidia, Server, ThinkSystem

Published: 2025-04-15

Length: 5 minutes

Start the training:

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: SXXW2545a

Related publications and links

For more information, see these resources:

- ThinkSystem SC777 V4 server product page
<https://www.lenovo.com/us/en/p/servers-storage/servers/supercomputing/lenovo-thinksystem-sc777-v4-neptune/len21ts0041>
- ThinkSystem N1380 enclosure product page
<https://www.lenovo.com/us/en/p/servers-storage/servers/supercomputing/lenovo-thinksystem-n1380-neptune/len21ts0038>
- ThinkSystem SC777 V4 datasheet
<https://lenovopress.lenovo.com/datasheet/ds0190-lenovo-thinksystem-sc777-v4>
- ThinkSystem N1380 enclosure datasheet
<https://lenovopress.com/DS0187>
- Interactive 3D Tour of the ThinkSystem SC777 V4:
<https://lenovopress.lenovo.com/lp2047-3d-tour-thinksystem-sc777-v4>
- ThinkSystem SC777 V4 drivers and support
<http://datacentersupport.lenovo.com/products/servers/thinksystem/sc777v4/7dmp/downloads>
- ServerProven hardware compatibility:
<https://serverproven.lenovo.com/>
- Data Center Solution Configurator (DCSC)
<https://dcsc.lenovo.com>
- Lenovo Cluster solutions configurator (x-config)
<https://lesc.lenovo.com/products/hardware/configurator/worldwide/bhui/asit/index.html>

Related product families

Product families related to this document are the following:

- [AI Servers](#)
- [Artificial Intelligence](#)
- [High Performance Computing](#)
- [Supercomputing Servers](#)
- [ThinkSystem SC777 V4 server](#)

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